From Feudalism to Agricultural Capitalism: An agent-based model of the Brenner-Woods theory of transition

Abstract

The results of the Brenner Debate produced an enhanced theory of transition from feudalism to capitalism based in rural 17^{th} Century England. A key tenant of this school of thought is the role of "improvement" as both a motor in the divorce between peasants and their means of subsistence, and a consequence of newly created market imperatives both landlords and tenants were now subject too. To test the role of improvement ideology in the transition an agent based model is constructed. This incorporates tenants, landlords, ecological deterioration and rental markets. The critical role of rental market pressure is modelled through a transition from customary rent to leasehold rents.

1 Introduction

According to [Comninel, 2000, p.10] feudalism may be defined as a mode of production in which the ruling class extracts surplus value from direct producers via non-economic means, often political and under the threat of force. In contrast, capitalism extracts surplus value from labourers, who are not direct producers and do not posses the means of production, via economic means. Capitalists must re-invest the extracted surplus value, in order to reproduce their class position, to keep up with increasing productivity. Through an analysis of the transition from one class structure to another, one might hope to gain insights into common identifiers which are associated with the collapse of a mode of production. Improvement ideology emerged in the mid 16^{th} Century composed of both practical methods to increase the output of a given piece of land, and as a method through which to better the condition of the English peoples, as argued by Captain Walter Blith.

1.1 Brenner's theory of transition

With the publication of "Agrarian Class Structure and Economic Development in Pre-Industrial Europe" in 1976 Robert Brenner produced a crucial advance in the causes of the very first case study of the transition from feudalism to capitalism. It is in this very word "transition" that Brenner argued even Marxist historians such as Paul Sweezy or Maurice Dobb had been led astray [Brenner, 1977, p.41-53], emphasising that they were assuming an a priori existence of an embryonic capitalism within feudal society.Several theories of transition may be grouped as commercial models. The keystone of these theories is that capitalism arose as the barriers for its development were removed [Wood, 2002, p.12]. Crucially, the driving factor behind this development was man's "propensity to truck, barter, and exchange" [Smith and Stewart, 1963]. This force led to the division of labour and ultimately industrial capitalism. Brenner and Wood's fundamental critique of this approach is that it assumes a nascent agricultural or industrial capitalism within other modes of production, such that there is no real transition and capitalism was inevitable [Wood, 2002, p.51]. Brenner's theory sought internal conflicts within feudalism, as opposed to an external impetus, which might produce its collapse. Wood offers an analysis that " it [was] a matter of lords and peasants, in certain specific conditions peculiar to England, involuntarily setting in train a capitalist dynamic while acting in class conflict with each other to reproduce themselves as they were" [Wood, 2002, p.52]. This focus on the class conflict nature of the transition is what distinguished Brenner and Wood's theory from other contemporary Political Marxists. Brenner identifies the internal logic of feudalism that inhibits surplus reinvestment. For the feudal lord it was often easier to find new avenues to "squeeze" [Brenner, 1976, p.74] out more absolute surplus value from their subjects instead of increasing their labour efficiency. Due to the fact that a lord's power was directly correlated to size of the population under his control this lead to a focus on labour immobility, and maximising the ease with which he could extract surplus value from his peasants by non-economic means. Resulting in restrictions on land mobility and "unfree peasants [unable] to convey their land to other peasants without the lord's permission" [Brenner, 1976, p.50]. The centuries previous to the 17^{th} Century were characterised by the end of serfdom in England, resulting in peasants whose lords extracted surplus value partly through the rent of their land. These were customary rents that decreased in value for the lord over time due to inflation. This loss in income revenue was counteracted through the imposition of arbitrary fines on peasants wherever land was sold or inherited. However over the 15^{th} and 16^{th} centuries landlords successfully crushed peasant land rights in order to combat this decreasing power. This decay in freehold rights meant that lord could slowly accumulate large demesne holdings, which could be leased out to larger tenants [Hoyle, 1990, p.7]. Who in turn sub-let to smaller farmers or employed wage labourers. This new wave of larger tenancies brought two new sources of market imperatives, firstly the lord had to keep his rents competitive as he was now trying to undercut the rest of the domestic tenancy market, given that the larger tenant had the ability to move away.[1] Secondly the tenant had to compete on a goods market, requiring them to sell the products of their husbandry in order to make enough profit to fulfil market dependant rent. This required a shift from production for subsistence to production for sale. These two forces resulted in a new weakly symbiotic relation between lord and capitalist tenant in which both were driven to re-invest surplus value back into their land, thereby improving agricultural yields and transforming feudal wealth into capital. This established a new set of laws of motion in which the lord and tenant were compelled to improve their agricultural yields in order to self-replicate their class condition. Brenner claims that "agricultural improvement not only made it possible for an ever greater proportion of the population to leave the land to enter industry" [Brenner, 1976, p.67] but also mean that a larger proportion of the domestic population was no longer involved in agriculture, freeing peasants to become the wage labour required for "England's continued industrial growth" [Brenner, 1976, p.67] through the 17^{th} Century.

2 Where the debate now stands

Before the questions, a word on the terrain, because it has moved. The exchange that followed Brenner's 1976 intervention [Aston and Philpin, 1985] was fought against demographic and commercial accounts, and §1.1 above reconstructs it on those terms. Neither is where the argument now is. Two later fronts have opened, and it is against these, not against Postan [Postan and Hatcher, 1978], that a contribution has to be positioned.

The first is *empirical*, and concerns whether English custom in fact decayed the way Brenner says. Whittle's Norfolk evidence finds an active peasant land market and customary tenure considerably more secure than the account requires, with capitalist relations emerging without wholesale dispossession [Whittle, 2000]; Bailey relocates the decline of serfdom in law and institutional change rather than in class struggle [Bailey, 2014]; Dyer makes the transition longer and earlier than the sixteenth–seventeenth-century window [Dyer, 2005]; Shaw-Taylor's quantitative work pushes proletarianisation later and reduces enclosure's share of it [Shaw-Taylor, 2012]; and Allen locates the productivity gains in yeomen and owner-occupiers rather than in large capitalist tenants on consolidated farms [Allen, 1992] – which, if right, misplaces the causal core of the improvement story. Dimmock's defence engages this evidence directly [Dimmock, 2014], and the collection edited by Whittle is the collective stocktake on tenure [Whittle, 2013].

The second is *methodological*, and concerns whether England may be explained from inside England at all. Anievas and Nişancıoğlu's charge is that Brenner and Wood are internalist, and thereby Eurocentric [Anievas and Nişancıoğlu, 2015]; Heller presses Anglocentrism and a case for French capitalism [Heller, 2011]; Banaji rejects the rigid mode-of-production boundary the argument needs [Banaji, 2010]; van Bavel's Low Countries arrived without the English tenurial structure at all [van Bavel, 2010]. The current Political Marxist answer is the programme set out by Lafrance and Post [Lafrance and Post, 2019], with Lafrance's France [Lafrance, 2019] the counterpart the model's own France regime (§4.7) should be read against rather than Brenner's 1976 sketch. Bois's original jibe, that this is *political* Marxism because the politics carries the explanation [Bois, 1978], is the ancestor of a line pressed since by Heller and by Davidson [Davidson, 2012];

[1] **Flagged for revision.** This sentence has the competitive pressure running from lords bidding against one another to retain mobile tenants. That is Postan's supply-and-demand account, which Brenner cites only in order to reject it: "the lords and the employers found that the most effective way of retaining labour was to pay higher wages, just as the most effective way of retaining tenants was to lower rents and release servile obligations" [Brenner, 1976, p.39, n.17, quoting Postan]. For the sixteenth century Brenner has the pressure running the other way, since population was rising: "competition for land induces the peasantry to accept a serious degradation of their personal/tenurial status in order to hold on to their land" [Brenner, 1976, p.38], and lords "could always get replacements, quite often indeed on better terms" [Brenner, 1976, p.44]. The market imperative on the lord's side is better stated as the opportunity to let to the highest bidder rather than the necessity of undercutting rivals – see §4.8.

RQ3 below is its direct test. Moore's complaint, that the account brackets nature [Moore, 2003, Moore, 2015], is the one this model is unusually well placed to answer rather than concede, since it has an ecological driver Brenner does not (§4.5).

The model is exposed on the second front in a way worth conceding at the outset: a closed, England-shaped lattice (§4.4) is methodological internalism implemented in code. The defence is not that the charge does not apply but that it applies more usefully here than to prose, since an assumption written as a boundary condition can be stated, varied and reported, where the same assumption carried as a habit of narration cannot. No question below tests it, and §6 returns to what that costs.

3 Research Questions

The model is built to ask one question: are the class-conflict mechanisms Brenner and Wood identify sufficient, on their own, to produce the outcomes the theory is invoked to explain? Six sub-questions divide that up, and each is posed against a named rival account rather than as a matter of reproducing a known pattern. This is deliberate. A question of the form "does the model reproduce X" puts nothing at risk, because for most X both answers are compatible with the theory; the form used instead is "the account commits us to X, and here is whether X survives". A second and separate set of questions (§**??**) asks whether the model behaves well enough for the first set to be reported at all. Keeping the two apart matters, because a modelling defect filed as a theoretical result is how a formalisation misleads.

3.1 Claims about the theory

RQ1: Does the improvement dynamic require competitive tenancy, or only market exposure under secure property? Brenner's mechanism is a landlord–tenant symbiosis in which competitively let, insecure tenancy is what compels reinvestment, displacing "the traditionally antagonistic relationship in which landlord 'squeezing' undermined tenant initiative" [Brenner, 1976, p.65]. Allen's rival account locates the same productivity gains in yeomen and owner-occupiers – that is, in *secure property* rather than in competitive tenancy [Allen, 1992]. The model can separate the two because it carries both tenures at once: Freehold (§4.7) responds to the market while paying no rent, so improving disposition $\iota_j^T(t)$ and capital $k_j(t)$ can be compared for Leasehold against Freehold on the same land in the same run, and again within tenure by an event study around each parcel's own conversion. Brenner's claim predicts that Leasehold leads; Allen's predicts that Freehold does. How sharply the comparison can be drawn depends on how strongly Freeholders feel the competitive yardstick, which the shadow rent of §4.7 sets: with it on, Allen's outcome is favoured by construction, and with it off Brenner's is, since Freeholders then face no competitive test at all. The reported answer is therefore the interval between the two settings, and an interval spanning both outcomes is itself a result – that the model as specified cannot separate them.

RQ2: Do Brenner's own mechanisms generate a spreading transition, or only a simultaneous one? Every mechanism Brenner names is uniform across England: inflation erodes fixed customary rents everywhere at the same rate (§4.5), the entry-fine loophole is open to every lord [Brenner, 1976, p.62], and the state–landlord alliance is a national condition [Brenner, 1976, p.71-72]. Nothing in the stated account therefore explains why the transition should begin anywhere in particular, yet a transition occurring everywhere at once is neither the historical pattern nor what the theory is usually taken to describe. The test is the distance-dependent conversion lag of §4.17(iii), measured with the two transmission channels of §4.8 ablated. If removing them yields simultaneity, the account is incomplete as stated and requires an inter-estate mechanism Brenner never specifies. That is a constructive result rather than a refutation: it identifies what the theory needs and does not have, which is the kind of gap a formalisation can expose and prose can leave unasked.

RQ3: Is the state–landlord alliance necessary, sufficient, or merely permissive? Bois named this school *political* Marxism as an accusation: that the politics carries the explanation while class conflict decorates it [Bois, 1978], a charge pressed since by Heller [Heller, 2011] and Davidson [Davidson, 2012]. The model can test it because the two are separable – institutional resistance θ (§4.6) sets the alignment of Crown and gentry, and the class-conflict mechanism can be disabled independently, as $\alpha_1 = 0$ so that a lord's fiscal pressure no longer drives conversion, together with $\varphi = 0$ so that conversion no longer competes against the squeeze (§4.10). A single England/France flip would show only that θ matters, which nobody disputes; the test is the 2×2 of both θ regimes crossed with the mechanism enabled and disabled. If the English outcome survives with class conflict disabled, the politics is sufficient on its own and Bois was right; if θ merely widens or narrows a transition that class conflict drives either way, it is permissive and the charge fails. The France arm is read against Lafrance [Lafrance, 2019], for whom France has a peasant-proprietor dynamic of its own rather than a mere absence of England's – which is what the Freehold pathway of §4.7 represents.

RQ4: How secure could customary tenure have been and still have given way? Whittle finds custom more secure, and the peasant land market more active, than Brenner's account allows [Whittle, 2000]; Bailey attributes its decline to law rather than to struggle [Bailey, 2014]; Dimmock disputes both [Dimmock, 2014]. What none of them can establish is how much security is *compatible* with the English outcome, because that is a counterfactual and the evidence is a single realised history. Sweeping the hazard of reopening terms ϖ, the arbitrary-fine rate φ and institutional resistance θ locates the frontier beyond which the transition no longer completes within the run. The result is a bound rather than a verdict – custom could have been this secure and England would still have arrived, but no more secure than this – and it answers a question the archive cannot be asked.

RQ5: Is dispossession necessary to the transition, or only sufficient? Whittle's stronger claim is that agrarian capitalism developed in Norfolk without the wholesale dispossession Brenner's narrative foregrounds [Whittle, 2000]. The model distinguishes the routes by which a parcel changes hands (§4.13): at a succession vacancy an heir is displaced only by an entry fine, whereas at a crisis vacancy no claimant remains to contest the holding. Damping the crisis channel, $\eta_1 = \eta_2 = 0$ with eviction suspended, leaves conversion to run through succession and fine escalation alone. If Leasehold dominance, farm concentration and a wage-labour pool still emerge, then dispossession is sufficient but not necessary, which reconciles Brenner's logic with Whittle's evidence rather than forcing a choice between them; if they do not, Brenner's emphasis is vindicated on his own terms.

RQ6: Is ecology doing explanatory work the theory attributes to class structure? The model has an exogenous ecological driver (§4.5) and a real spatial distribution of land quality (§4.4), neither of which Brenner has. Moore's charge is that the account brackets nature [Moore, 2003, Moore, 2015], so if the timing and location of the transition here are set by soil rather than by class position, the model establishes Moore's point using Brenner's own mechanisms. The ablation is `uniform_fertility`, which flattens the ALC-derived regional structure to its national mean while retaining parcel-level jitter, so that land quality loses its spatial *pattern* without losing its heterogeneity. Because the distance regressor still measures from the same county, a lag surviving the ablation is by construction not an ecological one. To our knowledge this comparison has not previously been made.

4 Model Design

(GENERALLY THE TEXT READS FAR TO MATHEMATICAL, A HISTORIAN NEEDS TO BE ABLE TO READ THIS SO THE DESCIRIPTION OF EQUATIONS AND REALTIONSHIPS SHOUDL COME FIRST WITH CITATIONS ETC THEN ITS REPRESENTATION IN EQUATION FORM AS A BONUS FOR THOSE FOR QUANTITATIVE.) The model is designed as a test of the internal consistency of the Brenner-Wood account. The question it asks is whether the class-conflict mechanisms Brenner and Wood identify are jointly sufficient to generate, from a small and spatially localised seed, the macroscopic outcomes the theory is invoked to explain: the collapse of customary tenure, the concentration of landholding, the emergence of the landlord/capitalist-tenant/wage-labourer triad [Brenner, 1976, p.63], and the spread of an "improvement" ethic that legitimises enclosure [Wood, 2002, p.108-109].

4.1 Causal structure

(NEED A LIST OF THE DYANMICS AND STYLIZED FACTS REPRESENTED BEFORE STARTING THIS SECTION AS ITS VERY LONG AND KIND CONFUSING) Figure 1 summarises how the mechanisms below connect; refs for each node are given in the corresponding subsection of the text. Two features are worth flagging before the equations.

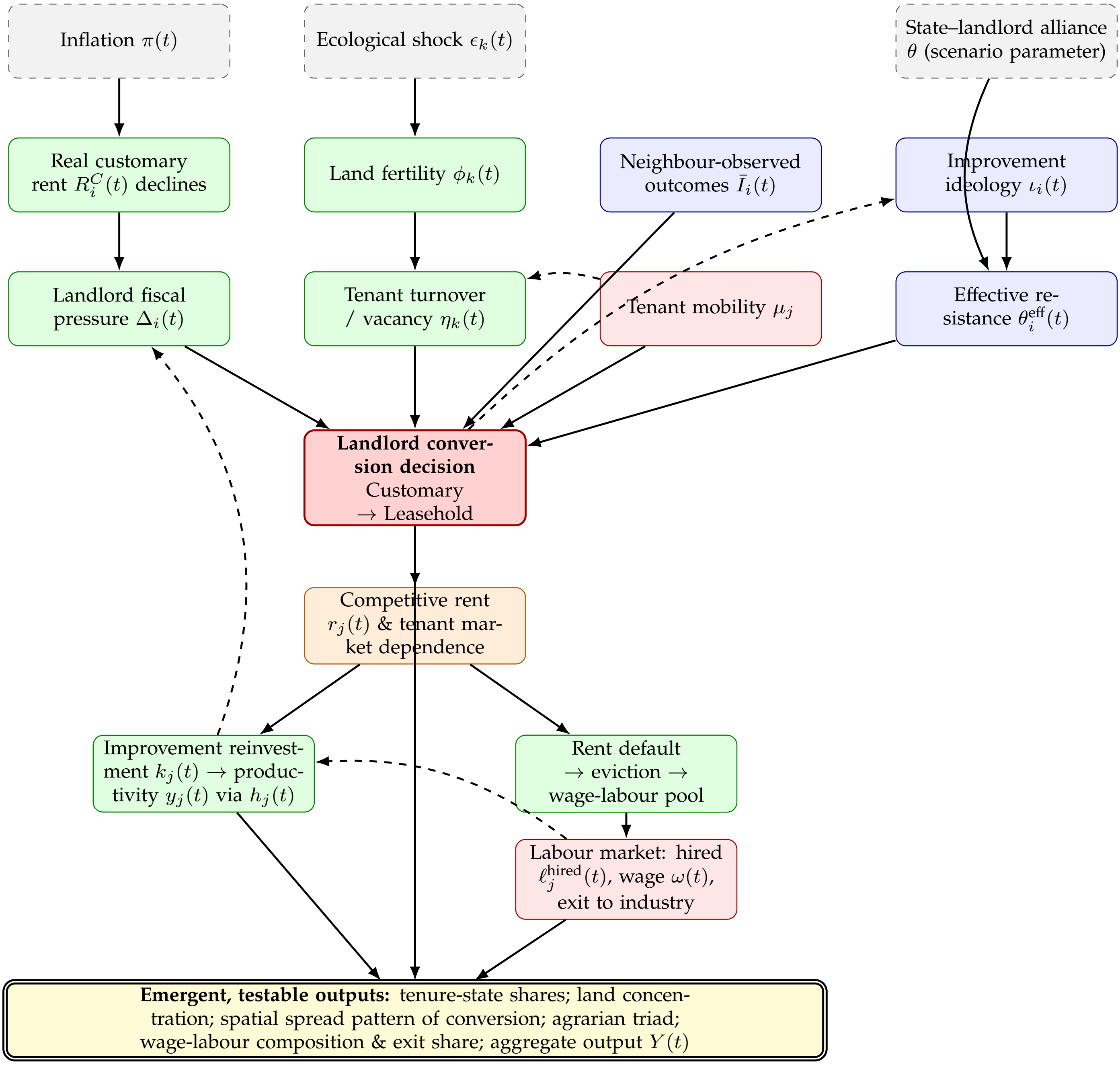

Figure 1: (UPDTE FIGURE CHECK ITS LATEST GIVEN EQUATIONS, INCLDUE Freehold entrenchment adn enclosure of the commons, DONT REFRENCE SECTION INSTEAD DESCRIBE WHAT IT IS)Causal structure of the model. Dashed boxes are exogenous/scenario parameters; solid green boxes are endogenous processes; blue boxes are the three spread channels of §4.8; the red box is the landlord's discrete decision; the double yellow box lists the emergent quantities used as validation targets in §4.17. Dashed arrows denote feedback loops.

4.2 Model schedule

(NEED TO MAKE SURE ITS UPDATED) The mechanisms above are specified as update rules for individual state variables; what has not yet been fixed is the order in which they run within a single period $t \to t+1$, nor the bookkeeping state – beyond tenure, holdings, wealth and capital – needed to evaluate them. Table 1 lists every piece of state actually required, including the consecutive-shortfall counter $d_j(t)$ that the τ-period eviction rule (§4.10) presupposes but does not itself name. The list below then fixes the order. All quantities dated t (rents, fiscal pressure, the observed-outcomes and ideology terms, wage) are computed once at the start of the period from state as of t and held fixed while every within-period event is resolved – a standard synchronous-update convention, adopted so that two vacancies arising on the same estate in the same period are resolved against the same $\Delta_i(t)$ rather than one seeing a value the other has already changed.

Agent	State	Meaning
Landlord i	$W_i(t)$, $\iota_i(t)$	wealth; legitimating ideology exposure (§4.8)
	$\mathcal{T}_i(t)$	set of currently active, separately-let tenancy slots on the estate
Tenant j	$\sigma_j(t)$, $\mathcal{P}_j(t)$	tenure state; set of held parcels
	$w_j(t)$, $k_j(t)$, $\iota_j^T(t)$	wealth; capital; improving disposition (§4.10)
	$n_j(t)$	household size in adult labour equivalents (§4.14)
	$c_{\ell,j}$	own landless subsistence requirement, drawn once (§4.15)
	$d_j(t)$	consecutive periods $w_j(t)$ has been short of $\rho_j(t)+c_j$; eviction at $d_j(t)=\tau$
Landless ℓ	$w_\ell(t)$, $n_\ell(t)$	wealth; size, supplying $n_\ell(t)\bar{\ell}$ to the labour pool
Urban u	$w_u(t)$, $n_u(t)$	wealth; size. Still a household: it earns, eats, reproduces and may return (§4.15)
Parcel k	$\phi_k(t)$, $\bar{\phi}_k$	current fertility stock; fixed carrying capacity (§4.5)
	commons-flag	fixed at $t=0$ (§4.12)
Global	$\omega(t)$, $P(t)$, $\Pi(t)$, $\Xi(t)$	wage; produce price; price index; enclosure share

Table 1: State carried by each agent type, in addition to the fixed per-agent parameters of Table 2.

1. **Exogenous update.** Draw $\pi(t)$ and update $\Pi(t)$; draw $\epsilon_k(t)$ for every parcel. Since $\phi_k(t+1)$ ((3)) depends only on $\phi_k(t)$ and $\epsilon_k(t)$, it may be computed at any point in this list. Update the produce price $P(t+1)$ against urban demand (§4.15).

2. **Labour demand and clearing.** For every Leasehold or Freehold tenant, compute $\ell_j^{\text{demand}}(t)$ from $k_j(t)$ and $n_j(t)$; ration to $\ell_j^{\text{hired}}(t)$ against the labour supply $S(t)$; update $\omega(t+1)$ (§4.15).

3. **Production.** Compute $y_j(t)$ ((18)) for every occupied tenancy from $\Phi_j(t)$, $k_j(t)$ and $h_j(t) = n_j(t)\bar{\ell} + \ell_j^{\text{hired}}(t)$.

4. **Wealth, capital and disposition.** With $\rho_j(t)$ already fixed at the start of t (Customary and Leasehold rents were set at $t-1$; Freehold is always 0), update $w_j(t+1)$ ((13)), $M_j(t)$ and $\iota_j^T(t+1)$ ((15)), and $k_j(t+1)$; update $d_j(t+1)$ (reset to 0 if solvent, incremented otherwise). Landless households update $w_j(t+1)$ symmetrically and draw $p_j^{\text{exit}}(t)$ ((30)); urban households update theirs at $\omega^U - c^{\text{urban}}$ per member and draw $p_j^{\text{return}}(t)$ ((31)). Record every household's per-head surplus $g_j(t)$ ((22)) for the next step.

5. **Demography.** For every surviving household, urban ones included, draw deaths, then births, from (23) and update $n_j(t+1)$; mark households reaching $n_j = 0$ extinct; then draw partition ((25)) and create the resulting landless households (§4.14). Households created in this step take no part in it, so a household cannot be founded and reproduce in the same period.

6. **Landlord fiscal pressure.** Compute $R_i^C(t)$, $R_i^L(t) = |\mathcal{T}_i^L(t)|\,\hat{r}_i(t)$, and $\Delta_i(t)$ (§4.6).

7. **Spread terms.** Compute $\bar{I}_i(t)$ from $s_{i'}^L(t)$ and $W_{i'}(t)$ as of the start of t (§4.8); update landlord ideology $\iota_i(t+1)$ and effective resistance $\theta_i^{\text{eff}}(t)$; update enclosure share $\Xi(t+1)$ (§4.12).

8. **Vacancy determination.** For every occupied parcel, draw a succession event (probability η_0) or a crisis event (the excess-hazard terms of (21), or $d_j(t+1) = \tau$, or a successful migration draw $p_j^{\text{migrate}}(t)$, §4.8); collect the resulting list of vacancies (§4.13).

9. **Vacancy resolution.** Resolve every vacancy collected in step 7, in random order, by the fixed procedure of §4.13, using the period-t values of $\Delta_i(t)$, $q_i(t)$, $p_i^{\text{convert}}(t)$ and $p_j^{\text{Freehold}}(t)$ computed above – *not* recomputed between vacancies within the same period.

10. **Rent reset.** Recompute $\hat{y}_i(t)$ (§4.10) from this period's realised Leasehold output (step 3) over $\mathcal{T}_i^L(t)$ as it stood during production, giving $\hat{r}_i(t+1) = \theta_{\text{rent}}\hat{y}_i(t)$ for the next period.

11. Advance to $t+1$ with all updated state now current.

4.3 Agents

Landlord $i \in \{1, \ldots, N_L\}$. Each landlord holds an estate of tenancies $\mathcal{T}_i$, a wealth stock $W_i(t)$, and a fixed consumption requirement C_i tied to the reproduction of aristocratic class position rather than to profit-maximisation as such [Brenner, 1976, p.31]. Landlords do not choose to become capitalists; they act defensively, to avoid losing their own class position, and any systemic shift toward market dependence is a by-product of that defensive action, both lords and peasants "involuntarily setting in train a capitalist dynamic while acting in class conflict with each other to reproduce themselves as they were" [Wood, 2002, p.52]. Each landlord also has an awareness radius (NAME WHAT THE RADIUS VARIABLE IS) over neighbouring estates' rents and outcomes [Wood, 2002, p.100-101], and an exposure to "improvement" ideology $\iota_i(t)$.

Tenant household j. A tenant is a single agent type carrying a tenure state $\sigma_j(t) \in \{\text{Customary}, \text{Leasehold}, \text{Freehold}$ that evolves endogenously, and a holding $\mathcal{P}_j(t)$ – a set of land parcels, since holdings themselves grow and shrink through engrossment (§4.11) rather than being fixed at one parcel per tenant. This reflects Brenner and Wood's own emphasis that the triad is an *outcome* of class conflict, not a precondition of it [Brenner, 1976, p.63]; Freehold is included as the historical alternative outcome of the same conflict that prevailed in France rather than England (§4.7).

The agent is a *household* rather than a person, and carries a size $n_j(t) \in \{1, 2, \ldots\}$ in adult labour equivalents. It supplies $n_j(t)\bar{\ell}$ labour to its own holding and owes $n_j(t)\, c^{\text{head}}$ in subsistence, so a member is simultaneously a pair of hands and a mouth; $n_j(t)$ then evolves through the fertility, mortality and partition rules of §4.14. Two things follow that a single-body agent cannot represent. First, the peasant family farm's characteristic response to pressure – absorbing more labour on the same land rather than buying it in – becomes available to the model, and because only *hired* labour is charged at $\omega(t)$ in (13) while family labour is not, the family farm has a genuine cost advantage over an equally sized holding worked by wage labour [Chayanov, 1966]. Second, it makes proletarianisation something a household can *produce* – the member for whom the holding has no profitable use, historically the non-inheriting son – and not merely something eviction does to it.

Land parcel k, held by a tenant, with a fertility/productivity state $\phi_k(t)$ subject to an exogenous ecological-demographic process (§4.5). Land is not a decision-maker, but its local, heterogeneous condition is what gives the model's initial "seed" of competition a non-arbitrary spatial location.

4.4 Spatial structure: an England-shaped lattice

A lattice is laid over the real outline of England and its heterogeneity is drawn from a real, if necessarily modern, proxy for regional land quality – so that the model's "small, spatially localised seed" (§4) is a location the data itself supplies, not one the modeller plants by hand. θ, the scenario parameter that stands for a specifically English alignment of Crown and gentry that Brenner and Wood both treat as sui generis to England [Brenner, 1976, Wood, 2002, p.68-75], such that a comparison with France can be made (§4.6).

Construction Land parcels k occupy cells of a lattice laid over England's bounding box at resolution L: the longer axis of the box is divided into L cells and the shorter axis takes as many square cells of that same size as it holds, so that a lattice distance is proportional to a real distance along both axes. At $L = 155$ this is a 155×136 grid, of which 7,417 cells survive once those falling outside the country's boundary

[Office for National Statistics, 2022] are excluded outright. Each surviving cell is assigned, once, to whichever of England's 39 historic counties its centroid falls within, and inherits that county's fertility baseline $\bar{\phi}^{\text{county}}$ (CAN WE NOT GET MORE ACCURATE?), computed by area-weighting Natural England's Agricultural Land Classification (ALC) grades [Natural England, 2021] within the county (Grade 1, "excellent", mapped to the highest score, Grade 5, "very poor", to the lowest). Each parcel then draws

$$\bar{\phi}_k \sim \mathcal{U}\big(\bar{\phi}^{\text{county}(k)} - \zeta,\ \bar{\phi}^{\text{county}(k)} + \zeta\big), \tag{1}$$

retaining fine-grained local heterogeneity within a county while centring it on a real regional value: modern ALC grades reflect a soil-and-climate endowment that changes on a geological, not a century, timescale, so treating it as invariant over the model's ~150-year window is a stated proxy assumption, distinct from the fast-moving, endogenous depletion and regrowth (3) already gives each parcel. The model's low-fertility "seed" is then simply wherever this construction happens to put the worst county – historically the upland, wood-pasture and moorland counties rather than the arable South and East – with no separate seed-radius parameter required.

N_L landlord seed points are scattered uniformly at random within *each* historic county (uniform rather than weighted toward more heavily manorialised regions (CAN WE FIND THE INFO FOR THIS?). Estates are then formed by a Voronoi tessellation around these seeds *within each county*: parcel k belongs to landlord i's estate $\mathcal{T}_i(0)$ if i's seed is k's nearest seed among those falling in k's own county. Seeding per county rather than nationally is what makes the county a genuine upper tier rather than a label: no estate straddles a county boundary, so each estate inherits exactly one fertility baseline $\bar{\phi}^{\text{county}}$, and the ecological seed is an estate-level condition rather than one cutting across estates. This gives each landlord a contiguous but irregularly shaped estate without requiring the modeller to hand-draw manor boundaries.

Two distinct neighbour relations are read off the same lattice, since Brenner and Wood's own language keeps them at different levels: engrossment operates "within the individual tenancy relation" (§4.12), i.e. between adjacent *parcels* under a single landlord, whereas the spread channels of §4.8 (NAME THIS NOT SECTION) operate between *estates*.

- **Parcel adjacency** $\mathcal{N}(k)$ (§4.11): the Moore (8-connected) lattice neighbours of k that belong to the same landlord's estate as k.
- **Landlord awareness radius** $\mathcal{N}_i$ (§4.6, §4.8): all landlords $i' \neq i$ whose seed point lies within Euclidean distance r of i's seed point.

4.5 Exogenous drivers: why feudalism cannot reproduce itself

Two exogenous processes represent the standing pressure that makes the customary order unsustainable on its own terms, independent of any competitive dynamic.

Inflation. Customary rent $r_j^C(0)$ is fixed in nominal terms at the point a tenancy is granted and is not renegotiated while a tenant remains in the Customary state: "rent *per se* (*redditus*) was fixed by custom, and subject to declining long-term value in the face of inflation" [Brenner, 1976, p.61]. With an exogenous price index $\Pi(t) = \prod_{s=1}^{t}(1 + \pi(s))$, the real value of landlord i's customary income is

$$R_i^C(t) = \sum_{j \in \mathcal{T}_i^C(t)} \sum_{k \in \mathcal{P}_j(t)} \frac{r_k^C(0)}{\Pi(t)}, \tag{2}$$

which decays monotonically even though nothing in the local land market has changed.

Ecological-demographic cycle. Land fertility is modelled as a renewable stock subject to logistic regrowth toward a parcel-specific carrying capacity $\bar{\phi}_k$ – the standard formalism for a resource that grows fastest at intermediate stock levels and saturates as it approaches its ecological ceiling [Verhulst, 1838], here combined with an extraction/shock term exactly as in the bioeconomic model of a harvested renewable resource [Schaefer, 1954] – punctuated by exogenous shocks $\epsilon_k(t)$:

$$\phi_k(t+1) = \phi_k(t) + \rho\, \phi_k(t)\Big(1 - \frac{\phi_k(t)}{\bar{\phi}_k}\Big) - \epsilon_k(t). \tag{3}$$

Neither Brenner nor Wood specifies a functional form for soil-fertility dynamics – nor, on inspection, does Jason W. Moore's account of feudalism's own ecological contradictions [Moore, 2003], which is historical-theoretical rather than mathematical. Their evidence is qualitative, concerning the exhaustion of arable land under insufficient manuring and the extension of cultivation onto marginal soils, and it is this qualitative claim, not a specific equation, that (3) is chosen to instantiate: the lord's surplus extraction "tended to confiscate not merely the peasant's income above subsistence...but at the same time threatened the funds necessary to refurbish...productivity" [Brenner, 1976, p.48], producing a "vicious cycle of the destruction of the peasants' means of support" in which "the crisis of productivity led to demographic crisis" [Brenner, 1976, p.49-50]. The heterogeneity of $\bar{\phi}_k$ across parcels is what gives the model's initial shock a location: some estates are ecologically buffered and some are not, mirroring this uneven pattern of subsistence crisis. A tenancy's turnover hazard rises when output falls below a subsistence threshold $\bar{y}$ (NOT CLEAR WHAT THE UNI OF THESE ARE OR HOW THEY AFFECT THE MODEL),

$$\eta_j^{\text{crisis}}(t) = \eta_1 \max\big(0,\ \bar{y} - y_j(t)\big), \qquad \eta_j^{\text{succ}}(t) = \eta_0, \tag{4}$$

drawn as two separate events on the *holding* rather than as one combined hazard on the parcel, because the two resolve differently at a vacancy (§4.13): a succession leaves a customary claimant, a crisis does not. The unit being the household's whole holding, $y_j(t)$ is its total output, so a holding that has engrossed its way well clear of $\bar{y}$ carries no crisis hazard from this term at all. Rent-default eviction, $d_j(t) = \tau$, takes precedence over both, and a crisis over a succession. So vacancies – through death, flight, or failure to hold the tenancy – cluster where ecological conditions are worst, exactly as Brenner describes for the fourteenth–fifteenth century demographic collapse: "the demographic collapse...left vacant many former customary peasant holdings. It appears often to have been possible for the landlord simply to appropriate these and add them to his demesne" [Brenner, 1976, p.61].

4.6 The landlord's conversion decision

At each vacancy, landlord i chooses between renewing the tenancy on customary terms or converting it to a leasehold let at a competitive rent. Define fiscal pressure as the gap between the landlord's consumption requirement and realised receipts, expressed as a share of that requirement,

$$\Delta_i(t) = \frac{C_i - R_i^C(t) - R_i^L(t) - F_i(t)}{C_i}, \tag{5}$$

where $F_i(t)$ is the arbitrary-fine income of §4.10. Two features of this deserve comment, because the obvious alternative is misleading. First, pressure enters as a *relative* shortfall: an absolute gap would mix a per-landlord requirement against rent summed over the estate, so a small estate would register far more pressure than a large one for purely arithmetic reasons, and estate size rather than any mechanism in the theory would determine who converted first. Second, C_i is itself set as a multiple of the estate's own initial rent roll rather than drawn independently of it. Drawn independently, every lord begins in heavy deficit and conversion is already likely everywhere in the first period, which leaves no interval during which custom is sustainable and therefore nothing for a localised shock to break – the dynamic §4.5 is built to produce. Scaling the requirement to the estate makes $\Delta_i(0) \approx 0$, so that fiscal pressure is genuinely *created* by the inflationary erosion of a fixed nominal rent.

The conversion probability is (STATE ITS A LOGISTIC FUNCTION EARLIER THAN CURRENLTY)

$$p_i^{\text{convert}}(t) = \sigma\Big(\alpha_0 + \alpha_1 \Delta_i(t) + \alpha_2 \bar{I}_i(t) - \alpha_3\, \theta_i^{\text{eff}}(t)\Big), \tag{6}$$

with α_0 and α_1 calibrated (HOW ARE THESE CALIBRATED) so that complete erosion of customary income is *necessary but not sufficient*: a lord with no converted neighbours still converts only rarely, and it takes the neighbour term $\alpha_2 \bar{I}_i(t)$ to carry the decision. This is not a free choice of parameters but a restatement of the paper's own claim in §4.8 – that uniform inflationary pressure cannot by itself generate a spreading rather than a simultaneous pattern. Coefficients that let $\alpha_1 \Delta_i(t)$ saturate the logit on its own would assume away the question RQ2 asks. with $\sigma(\cdot)$ the logistic function, $\bar{I}_i(t)$ the neighbour-observation term (§4.8), and $\theta_i^{\text{eff}}(t)$ the effective strength of institutional resistance a tenant can muster against conversion (§4.8). This is also the point at which landlords historically had a second, non-confrontational lever besides outright conversion: raising the entry fine charged at conveyance, since "there often remained one crucial loophole open to the landlord...He could insist on the right to charge fines...whenever peasant land was conveyed...in the end entry fines appear to have provided the landlords with the lever they needed to dispose of customary peasant tenants, for in the long run fines could be substituted for competitive commercial rents" [Brenner, 1976, p.62]; the model treats a fine escalation as a special case of the same conversion decision, since its economic effect – a rent that tracks the market rather than custom – is identical.

Conversion in place. That last point has a consequence the vacancy framing obscures. If conversion is only ever reached *at* a vacancy, and the incoming occupant is drawn from the landless pool (§4.13), then scarcely any tenant is present both before and after the conversion of their own holding – and the central question of §3, whether market exposure changes a tenant's disposition, becomes unaskable, since what the data would record is a change of occupant. But the entry fine was levied on the sitting tenant, and it is precisely by that route that "in the long run fines could be substituted for competitive commercial rents" [Brenner, 1976, p.62]. The model therefore also lets a lord reopen terms mid-tenancy: with per-period hazard ϖ the decision above is applied to a sitting customary tenant, who either meets the fine $\chi\,\hat{r}_i(t)|\mathcal{P}_j(t)|$ and continues as a leaseholder, or cannot and is dispossessed, the holding passing to the market. Conversion by re-letting and conversion by fine escalation are the same decision reached by two routes, and only the second leaves a tenant to observe. (IM NOT CONVINCED BY THIS MECHANISM OF MID RENTAL CONTRACT FINE, IT SHOULD BE ON SUCCESSION I THINK THATS WHAT THE SOURCES STATE, WHAT IS YOUR EVIDENCE FOR IT, IF NONE THEN REMOVE.)

Crucially, θ is not derived endogenously from a submodel of peasant revolt. Brenner's own comparative argument treats the presence or absence of an independent political ally for the peasantry as the decisive, largely exogenous condition distinguishing England from France: in France "the monarchy...had an interest in limiting the landlords' rents...and thus in intervening against the landlords to end peasant unfreedom and to establish and secure peasant property" [Brenner, 1976, p.69], whereas in England the same centralising process ran through, not against, the landlord class – "important sections of the English nobility and gentry were willing to support the monarchy's centralizing political battle against the disruptive activities of the magnate-warlords...but it was precisely these same landlord elements who were most concerned to undermine peasant property" [Brenner, 1976, p.71-72]. Wood's account of the English state makes the same point institutionally: the aristocracy was "demilitarized before any other aristocracy in Europe...part of the increasingly centralized state...[but] did not possess autonomous 'extra-economic' powers...to the same degree as their continental counterparts" [Wood, 2002, p.99], so that landlords "depended less on their ability to squeeze more rents out of their tenants by direct, coercive means than on their tenants' success in competitive production" [Wood, 2002, p.100]. Accordingly θ is set as a scenario-level parameter – an "England" regime and a counterfactual "France" regime, used for validation in §4.17 – rather than as an emergent variable, with individual conversion attempts still subject to idiosyncratic stochastic resistance through the logistic error term, representing the real but ultimately unsuccessful local resistance of the sort seen in Ket's Rebellion of 1549: "if successful, the peasant revolts of the sixteenth century...might have 'clipped the wings of rural capitalism'...But they did not succeed" [Brenner, 1976, p.62-63].

4.7 Freehold tenure and the closure of an alternative path

The conversion decision above is not the only possible outcome of a customary tenant's struggle against a landlord under fiscal pressure. In the mid-fifteenth century, before landlords found the entry-fine loophole, the outcome was genuinely open: "peasant tenants at this time were striving hard for full and essentially freehold control over their customary tenements, and were not far from achieving it" [Brenner, 1976, p.61]. Freehold is therefore included as a fourth tenure state, not merely as a residual category, because the theory's comparative logic requires a place for the path England did *not* take.

Initial condition, not emergent outcome. Consistent with Wood's account, the initial share of Freehold tenants is set low and is not itself a target the England-regime run is expected to explain: "the concentration of English landholding meant that an unusually large proportion of land was worked not by peasant-proprietors but by tenants...This was true even before the waves of dispossession...associated with 'enclosure"' [Wood, 2002, p.99-100], in contrast to France, which "remained a country of peasant proprietors" while "land in England was concentrated in far fewer hands" [Wood, 2002, p.133].

Freeholders still compete. Owning land outright does not exempt a tenant from the market imperative, since the compulsion runs through the goods market as much as the rental market: "yeomen were typically the very kind of capitalist tenants who were subject to competitive pressures, and even owner-occupiers would be subject to those pressures once the competitive productivity of agrarian capitalism set the terms of economic survival" [Wood, 2002, p.54]. A Freehold tenant's wealth therefore evolves by the same unified rule as any other tenure state ((13), §4.10, with $\rho_j(t) = 0$), using the same production function (18) and the same competitively-driven improving disposition $\iota_j^T(t)$ and reinvestment rate $\bar{s}\,\iota_j^T(t)$ (§4.10, (15)), with the estate's leasehold benchmark $\hat{r}_i(t)$ standing in as j's shadow rent for the purposes of $M_j(t)$ even though none is actually paid; a Freeholder who cannot sustain c_j for τ periods sells up and enters the Landless state, while one who prospers can absorb a neighbouring vacant parcel through the same engrossment competition available to capitalist tenants (§4.11) – "the same would increasingly be true even of landowners working their own land. In this competitive environment, productive farmers prospered and their holdings were likely to grow, while less competitive producers went to the wall and joined the propertyless classes" [Wood, 2002, p.103]. (HOW DOES A FREEHOLD LOSE MONEY? DOES THEIR CONSUMPTION INCREASE OVER TIME, IS THIS NOT AN ATTRACTOR STATE)

A rare, closing pathway. (WAS THIS PATHWAY REAL, WHERE IN THE TEXT IS THIS DESCRIBED, HOW FREQUENT WAS IT? FEELS LIKE ITS HERE TO CLOSE A LOOP NOT BECAUSE ITS REAL) At a vacancy where the sitting customary tenant's resistance succeeds (§4.6), there is a further, small probability that the outcome entrenches as Freehold rather than reverting to Customary,

$$p_j^{\text{Freehold}}(t) = \sigma\big(\lambda_0 + \lambda_1\,\theta - \lambda_2 \mathbb{1}[t \geq t^*]\big), \tag{7}$$

increasing in the same state–landlord alliance parameter θ used throughout, and falling discretely once the entry-fine loophole fully closes at the same t^* that marks the end of state resistance to enclosure (§4.12) – formalising the closing-off already asserted qualitatively below. In the England regime (θ low) this is close to zero even before t^*; in the France regime (θ high) this pathway dominates, giving the model's England/France comparison (§4.17) a mechanism rather than only a suppressed probability – the same underlying pressure resolves to the Leasehold triad in one regime and to entrenched Freehold smallholding in the other, exactly the divergence Brenner draws out at length in his own comparison of the two countries' seventeenth-century agrarian structures [Brenner, 1976, p.68-75].

4.8 Spread of the market imperative

Given uniform inflationary pressure, fiscal deficits accrue to all landlords roughly together; what produces a spreading, rather than simultaneous, pattern of conversion is that converting is a costly institutional break, so landlords are heterogeneously reluctant to be first. Two channels transmit a successful defection to its neighbours: what a lord can observe of a neighbour's results, and the slower diffusion of a legitimating ethic.

Why not tenant mobility. An earlier version of this model treated tenant mobility as a third transmission channel, on the reasoning that a tenant leaving for better terms elsewhere forces the losing lord's hand. That has to be withdrawn, for two reasons. The first is chronological: the fight over mobility is settled before the period modelled here begins. Controls on movement were constitutive of serfdom – their point was "preventing peasant mobility and thus a 'free market in tenants"' [Brenner, 1976, p.44] – but "by the mid-fifteenth century, through flight and resistance, [the peasantry was able] to break definitively feudal controls over its mobility and to win full freedom" [Brenner, 1976, p.61]. Mobility is therefore a standing precondition of the model, not a variable within it. The second reason is more serious: in the sixteenth century, with population rising, the competitive pressure runs the other way. "Competition for land induces the peasantry to accept a serious degradation of their personal/tenurial status in order to hold on to their land" [Brenner, 1976, p.38], and lords, far from bidding to retain tenants, "could always get replacements, quite often indeed on better terms" [Brenner, 1976, p.44]. Lords competing to keep mobile tenants is Postan's mechanism, which Brenner reproduces only to reject (§1.1); building it in would smuggle the demographic account into a model meant to test the class-structural one.

What the sources do describe at this point is not migration but *allocation*, and it is given its own subsection below (§4.9).

Observed outcomes. Landlords within i's awareness radius $\mathcal{N}_i$ update on the realised wealth gain of neighbours in proportion to how much of those neighbours' own estates have already converted – a landlord's estate converts tenancy-by-tenancy, not all at once (§4.13), so the signal is a continuous Leasehold share $s^L_{i'}(t) = |\mathcal{T}^L_{i'}(t)|/|\mathcal{T}_{i'}(t)|$ rather than a binary "has converted" flag,

$$\bar{I}_i(t) = \frac{1}{|\mathcal{N}_i|} \sum_{i' \in \mathcal{N}_i} s^L_{i'}(t) \cdot \frac{W_{i'}(t)}{W_{i'}(0)}, \tag{8}$$

with $t = 0$ the simulation's start, at which $W_{i'}(0) > 0$ by construction (§4.16), and $|\mathcal{T}_{i'}(t)|$ in $s^L_{i'}(t)$ counting only currently active, separately-let tenancy slots on estate i' – a parcel absorbed by engrossment (§4.11) has ceased to be its own tenancy and drops out of both numerator and denominator. This directly reflects the historical practice Wood describes of landlords' surveyors explicitly pricing the gap between fixed and competitive rent – "we can watch the development of a new mentality by observing the landlord's surveyor as he computes the rental value of land...against the actual rents being paid by customary tenants...[calculating] 'the annual value beyond rent' or 'value above the oulde [sic] rent"' [Wood, 2002, p.101].

Ideological legitimation. Adoption of "improvement" as a legitimating ethic diffuses across landlords as a Bass-type contagion process, with the contagion term driven by neighbours' realised *conversions* $s^L_{i'}(t)$ rather than by their professed ideology,

$$\iota_i(t+1) = \iota_i(t) + \beta_1\big(1 - \iota_i(t)\big) + \beta_2\, \bar{s}^L_{\mathcal{N}_i}(t)\big(1 - \iota_i(t)\big), \tag{9}$$

and with β_1, the autonomous term, kept small. Letting ideology diffuse from ideology alone would make neither term refer to any actual break from custom, so ι_i would climb toward unity on a schedule of its own whether or not a single tenancy had converted – an exogenous clock rather than a transmission channel, and one whose ablation in §4.17 would test nothing. Figure 1 already draws the arrow from the conversion decision back into ideology; this is the equation that implements it. and lowers the effective force of institutional resistance faced at conversion,

$$\theta^{\text{eff}}_i(t) = \theta \cdot \big(1 - \xi_0 - \xi_1 \iota_i(t)\big). \tag{10}$$

This is the mechanism Wood documents in detail: enclosure and the extinguishing of customary right were pursued not only economically but legally and rhetorically, since "between the sixteenth and eighteenth centuries, there was growing pressure to extinguish customary rights that interfered with capitalist accumulation...traditional conceptions of property had to be replaced by new, capitalist conceptions of property" [Wood, 2002, p.108], with "reasons of 'improvement'...cited systematically as the basis of title to property and as grounds for extinguishing traditional rights" [Wood, 2002, p.115], of which Locke's labour theory of property was the paradigm articulation [Wood, 2002, p.108-112].

Both channels are implemented as independent switches so that the paper's Results section can ask which, if either, is necessary to reproduce a spatially spreading – rather than simultaneous – pattern of conversion: the model's central internal-validity test.

4.9 Competitive allocation: letting to the highest bidder

Withdrawing mobility does not remove competition between tenants; it relocates it. What Wood describes is a market in which lords choose among rival takers rather than tenants choosing among rival lords:

> "landlords...would increasingly seek tenants who could produce competitively and pay good rents by increasing their own productivity. This meant that success would breed success, and competitive

farmers would have increasing access to even more land, while others lost access altogether...In a system of 'competitive rents', in which landlords, wherever possible, would effectively *lease land to the highest bidder, at whatever rent the market would bear*" [Wood, 2002, p.101].

This is an allocation rule, and the model implements it as one. When a market-exposed parcel k falls to be let (§4.13), the eligible takers bid what the holding is worth to them, which given (18) is a share of the output each could expect to raise on it:

$$b_j(k) = \theta_{\text{rent}}\, P(t)\cdot \hat{y}_j(k), \qquad \hat{y}_j(k) = \begin{cases} y_j(t) \,/\, |\mathcal{P}_j(t)| & j \text{ a sitting market-exposed neighbour,} \\ \phi_k(t)^{\mu}(k^{\text{trad}})^{\gamma}(n_j\bar{\ell})^{1-\mu-\gamma} & j \text{ from the Landless pool,} \end{cases} \tag{11}$$

struck in money, since a rent denominated in bushels would not be comparable with the income it is paid out of. A sitting neighbour therefore bids on *revealed* productivity – what they are actually getting per parcel already – while a landless entrant can only bid what an unimproved holding worked by its own family would yield. Household size enters the entrant's bid through its labour $n_j(t)\bar{\ell}$, so a larger landless household outbids a smaller one: the family-labour advantage, priced into the auction. Landless households differ only in wealth and in how many hands they bring, and both raise what they can bid, so it is enough to enter the best of each – the wealthiest and the largest. The parcel goes to the highest bidder who can also meet the entry fine $\chi\, b_j(k)$, and the winning bid becomes that parcel's rent: this is what "whatever rent the market would bear" means, and it makes the competitive rent an outcome of the auction rather than a parameter imposed on it. Adjacent tenants enter the auction only if the lord is willing to consolidate, which is the engrossment draw $q_i(t)$ of §4.11; the two are the same event seen from opposite sides, the lord's decision whether to break the holding up and the successful farmer's opportunity to absorb it.

The consequence is cumulative and one-directional, which is Wood's point. A tenant who has improved bids from a higher productivity, wins more land, and spreads the same capital over more parcels; a tenant who has not is outbid and eventually has nothing to bid with. Note that this is a mechanism of *concentration*, not of *transmission*: it governs who ends up farming, not how the break from custom travels between estates. It is therefore reported separately from the two spread channels above, and is not one of the switches §4.17 ablates when asking what produces contagion. (WHO CAN BID IS IT A GLOBAL MARKET OR JUST THE TENANTS IN A SINGLE COUNTY OR UNDER A SINGLE LORD?)

4.10 Tenant response: market dependence and improvement

Every occupied tenancy, whatever its tenure state, faces a state-dependent rent

$$\rho_j(t) = \begin{cases} \sum_{k\in\mathcal{P}_j(t)} r_k^C(0) \,/\, \Pi(t) & \sigma_j(t) = \text{Customary,} \\ \sum_{k\in\mathcal{P}_j(t)} r_k(t) & \sigma_j(t) = \text{Leasehold,} \\ 0 & \sigma_j(t) = \text{Freehold,} \end{cases} \tag{12}$$

the rent accruing *per parcel* rather than per tenancy, since a rate charged once per tenancy would let a tenant engross five parcels and pay what they paid for one – making concentration rent-free and over-incentivising the very outcome §4.11 is meant to explain. Each leasehold parcel carries its own rent $r_k(t)$, fixed at the bid that won it at the auction of §4.9 and reset only when the parcel is next let; where competitive allocation is switched off, every leasehold parcel on estate i instead carries the estate benchmark $\hat{r}_i(t)$, so that $\rho_j(t) = \hat{r}_i(t)\,|\mathcal{P}_j(t)|$.

Wealth then evolves by a single rule regardless of state,

$$w_j(t+1) = w_j(t) + R_j(t) - \rho_j(t) - n_j(t)\, c^{\text{head}} - \omega(t)\, \ell_j^{\text{hired}}(t) - F_j(t), \tag{13}$$

where the credited quantity is revenue rather than output, $R_j(t) = P(t)\, y_j(t)$ for $\sigma_j(t) \in \{\text{Leasehold}, \text{Freehold}\}$ and $R_j(t) = y_j(t)$ under custom: market-exposed holdings sell their produce at the going price $P(t)$ of §4.15, while customary production is consumed at home and so is not revalued by the urban market (WHY IS THIS THE CASE SHOULD THERE NOT BE SOME POSSIBILITY HERE OF EXTRA PRODUCTION FOR

SALE, POTENTIALLY A UTILITY FUNCITON HERE DETERMINING THE NUMBER OF HOURS WORKED AND LEISURE TO BALANCE THIS, SEEMS UNFAIR THAT CUSTOMARY RENT CANT TAKE EXCESS TO MARKET). Subsistence is owed *per head*, which is what makes household size a real constraint rather than a free source of labour: output rises with labour as $n_j^{1-\mu-\gamma}$ while this term rises as n_j, so every holding has a size beyond which it cannot feed its own household. (Under the single-body population rules a household is one member and pays the flat c_j of Table 2 instead.) (13) nests both the Leasehold and Freehold cases of earlier drafts as $\rho_j(t) = \hat{r}_i(t)$ and $\rho_j(t) = 0$ respectively. The Leasehold rent is not individually negotiated with the sitting tenant but set once for the whole estate at landlord i's local competitive benchmark – a single going rate a tenant has no part in setting, which is what makes it genuinely competitive rather than a personalised squeeze on whoever happens to be struggling,

$$\hat{y}_i(t) = \frac{1}{|\mathcal{T}_i^L(t)|} \sum_{j \in \mathcal{T}_i^L(t)} y_j(t), \qquad \hat{r}_i(t) = \theta_{\text{rent}}\, P(t)\, \hat{y}_i(t-1), \qquad \theta_{\text{rent}} \in (0,1), \tag{14}$$

(bootstrapped, at a landlord's first-ever conversion, from the estate's own customary rent scale in place of an as-yet-undefined $\hat{y}_i(t-1)$), and struck in money rather than in produce, for the same reason the bid of (11) is: the benchmark is compared against, and paid out of, revenue that has already been priced. It is the going rate against which every Leasehold tenant on estate i is measured, aggregating to the estate's leasehold receipts $R_i^L(t) = \sum_{j \in \mathcal{T}_i^L(t)} \rho_j(t)$ used in $\Delta_i(t)$ (§4.6); no inflation discount is needed here since $\hat{r}_i(t)$ already re-prices with current output, unlike $r_j^C(0)$, which is fixed in nominal terms at grant. This is exactly the calculation Wood attributes to the landlord's own surveyor (§4.8): $\hat{r}_i(t)$ is the "rental value of land...against the actual rents being paid" [Wood, 2002, p.101], made explicit as a number rather than left as a comparison the landlord alone draws.

Improvement as a response to competitive necessity, not a precondition of it. Rather than gating reinvestment on tenure state by fiat, the model makes a tenant's willingness to reinvest an endogenous "improving disposition" $\iota_j^T(t) \in [0,1]$ – the superscript distinguishing it from the landlord's legitimating ideology $\iota_i(t)$ of §4.8, a conceptually distinct variable that happens to answer the same abstract-level question (why does anyone start improving?) for the other class of agent.

What drives it is *competitive selection*, not the level of rent. This distinction is essential and easy to get wrong. The intuitive formalisation – letting the rent-to-output ratio $\rho_j(t)/y_j(t)$ do the work – inverts the theory, because a customary tenant's rent is fixed in nominal terms and so is a *lump sum*: they retain the whole of any extra output they coax from the land, whereas a leaseholder assessed at θ_{rent} of output faces a marginal claim on every additional bushel. On that reading custom would supply the stronger incentive to improve, which is the opposite of Brenner's argument. What actually distinguishes the market-dependent tenant is that they are measured against a rate somebody else sets, and lose the holding for falling short of it. The driver is therefore the estate's competitive benchmark relative to the tenant's own revenue, and it applies only to those tenures whose continued occupation is contingent on meeting it:(WHAT IS THIS M VARIABLE, NEED TO NAME THEM BEFORE THEY ARE STATED TO OFTEN REFERING TO A SECTION, EQUATION OR VARIABLE WITHOUT DESCIRBING IT MAKES IT VERY HARD TO FOLLOW)

$$M_j(t) = \begin{cases} 0 & \sigma_j(t) = \text{Customary}, \\ \dfrac{\hat{r}_i(t)\,|\mathcal{P}_j(t)|}{R_j(t)} & \sigma_j(t) \in \{\text{Leasehold}, \text{Freehold}\}, \end{cases} \qquad \iota_j^T(t+1) = (1-\varsigma)\,\iota_j^T(t) + \beta^T M_j(t)\big(1-\iota_j^T(t)\big), \tag{15}$$

both sides of $M_j(t)$ being in money, so that the price carried by $\hat{r}_i(t)$ and by $R_j(t)$ cancels and the pressure is a pure ratio. Here ς is a decay rate: a disposition is sustained by the pressure that produced it, and fades in its absence, so that improvement tracks present circumstances rather than being an acquired trait a dispossessed tenant carries permanently into a tenure that exerts no such pressure. Reinvestment then scales continuously in the disposition, against a depreciating stock,

$$k_j(t+1) = \max\Big(k^{\text{trad}},\ (1-\varrho_k)\,k_j(t) + \bar{s}\,\iota_j^T(t)\max\big(0,\ w_j(t+1) - w^{\min}\big)\Big), \tag{16}$$

with ϱ_k the depreciation rate. Depreciation matters for more than realism: without it an improvement, once made, is permanent and free to maintain, and the compulsion the theory describes is a one-off rather than a standing condition of continued tenancy.

The other half: why custom does not improve. A zero in (15) states that custom applies no competitive test, but says nothing about whether a customary tenant *could* improve if minded to. Brenner supplies the answer, and the paper has already quoted it (§4.5): seigneurial extraction "tended to confiscate not merely the peasant's income above subsistence...but at the same time threatened the funds necessary to refurbish...productivity" [Brenner, 1976, p.48]. The model implements that directly as an arbitrary fine at rate φ on customary income above subsistence,

$$F_j(t) = \varphi \max\big(0,\ y_j(t) - \rho_j(t) - c_j\big), \qquad \sigma_j(t) = \text{Customary}, \tag{17}$$

deducted from the tenant and credited to the lord's receipts. The two mechanisms are complementary rather than redundant: (15) removes the compulsion to improve under custom, and $F_j(t)$ removes the means. Crediting the fine to the lord also gives the model the choice the theory turns on – a lord under fiscal pressure may squeeze harder instead of converting, so conversion has to out-compete an alternative rather than being the only available response.(CAN YOU PRODUCE A STATE HERE WHERE THERE IS SUFFICIENT SQUEEZING, BUT GOOD RESISTANCE TO CONVERSION IN THE FRENCH SENSE?)

Because $\hat{r}_i(t)$ tracks the estate's competitive peers rather than any individual tenant's realised output, the reverse of (15) also holds: a tenant who does not raise $\iota_j^T(t)$ – and hence $k_j(t)$ and $y_j(t)$ – falls behind a rent they had no part in setting, and sustained shortfall triggers the same τ-period eviction rule as rent default generally (below), vacating the parcel for engrossment by a more successful neighbour (§4.11). Reinvestment is therefore not only individually adaptive under pressure but also the criterion by which the tenant population is selected – compete, in the sense of (15), or be outcompeted, in the sense of §4.13. This is the same market compulsion already used to justify Freeholders' continued exposure (§4.7): "even owner-occupiers would be subject to those pressures once the competitive productivity of agrarian capitalism set the terms of economic survival" [Wood, 2002, p.54].

This raises output through a diminishing-returns function of the capital and labour applied across the tenant's whole holding $\mathcal{P}_j(t)$, not a single fixed parcel,

$$y_j(t) = \Phi_j(t)^{\mu} \cdot k_j(t)^{\gamma} \cdot h_j(t)^{1-\mu-\gamma}, \qquad \mu, \gamma \in (0,1),\ \mu + \gamma < 1, \tag{18}$$

where $\Phi_j(t) = \sum_{k \in \mathcal{P}_j(t)} \phi_k(t)$ is the fertility-weighted size of the holding and $h_j(t) = \bar{\ell} + \ell_j^{\text{hired}}(t)$ is total labour applied: the tenant's own household endowment $\bar{\ell}$ plus any hired labour $\ell_j^{\text{hired}}(t)$ drawn from the landless pool (§4.15). How $\mathcal{P}_j(t)$ itself grows is the subject of §4.11; here it is enough that this is the mechanism, not merely an analogy, that Brenner identifies as distinguishing English agrarian development: the displacement of "the traditionally antagonistic relationship in which landlord 'squeezing' undermined tenant initiative, by an emergent landlord-tenant symbiosis which brought mutual co-operation in investment and improvement" [Brenner, 1976, p.65]. A tenant of any tenure state who cannot cover $\rho_j(t) + c_j$ for τ consecutive periods loses the tenancy and enters the Landless state – not to leave the model, but to become the labour supply the surviving tenants in (18) draw on, as §4.15 sets out; this is a crisis-type vacancy in the sense of §4.13, with no heir to fall back on, and is the mechanism by which a tenant who fails to develop $\iota_j^T(t)$ (above) is removed from the model.

4.11 Land concentration: the engrossment of holdings

Neither Brenner nor Wood treats landlord-level estate size as something the sixteenth–seventeenth century dynamics generate: Wood is explicit that big-landlord concentration was already in place before the story starts – "land in England had for a long time been unusually concentrated, with big landlords holding an unusually large proportion, in conditions that enabled them to use their property in new ways" [Wood, 2002, p.99]. Landlord estate size $|\mathcal{T}_i(0)|$ is therefore initialised as a heterogeneous but fixed starting condition (§4.3), not an emergent output. What the theory does explain endogenously is a different margin entirely: the concentration of *farms*, i.e. tenant holdings, as customary smallholdings are merged into fewer, larger units –

"with the peasants' failure to establish essentially freehold control over the land, the landlords were able to *engross, consolidate and enclose*, to create large farms and lease them to capitalist tenants who could afford to make capitalist investments" [Brenner, 1976, p.63],

a pattern Brenner also documents in the comparative section via Bowden: "under the stimulus of growing population, rising agricultural prices, and mounting land values...large estates were built up at the expense of small holdings" [Brenner, 1976, p.42].

At any vacated parcel k – whether from the ecological-turnover channel of §4.5 or a rent-default eviction (§4.15) – the landlord's decision is therefore not only *whether* to convert (§4.6) but *to whom* to let it: a new, separate tenant, or an existing neighbouring tenant's holding. Writing $\mathcal{N}(k)$ for the Leasehold or Freehold tenants already adjacent to k and $j^* = \arg\max_{j\in\mathcal{N}(k)} k_j(t)$ for the most capital-intensive of them, the probability of engrossment into j^* rather than separate re-letting is

$$q_i(t) = \sigma\big(\psi_0 + \psi_1 \Delta_i(t) + \psi_2\, k_{j^*}(t)\big), \tag{19}$$

reusing the landlord's fiscal pressure $\Delta_i(t)$ (§4.6) and the neighbour's own capital stock $k_j(t)$ (§4.10) rather than introducing new state. If engrossment occurs, $\mathcal{P}_{j^*}(t+1) = \mathcal{P}_{j^*}(t) \cup \{k\}$, directly operationalising Wood's point that "success would breed success, and competitive farmers would have increasing access to even more land, while others lost access altogether" [Wood, 2002, p.100-101]. If not, the parcel enters the ordinary conversion decision of §4.6 as before.

Because the same displaced smallholder who loses the parcel typically becomes the labour the engrossing tenant now needs to work the larger holding (§4.15), the model produces tenant-farm concentration and the wage-labour pool as two faces of a single event, rather than as separately assumed processes.

4.12 Enclosure of the commons

(NOT SURE ABOUT THIS SECTION, DONT LOVE EQ 21) Engrossment operates parcel by parcel, within the individual tenancy relation. Enclosure is a distinct, slower process that operates on shared subsistence resources external to any single tenancy: "there existed common lands, on which members of the community might have grazing rights or the right to collect firewood, and there were various other kinds of use rights on private land, such as the right to collect the leavings of the harvest during specified periods of the year" [Wood, 2002, p.107]. Enclosure is the extinguishing of exactly these rights, not merely the fencing of land: "enclosure meant not simply a physical fencing of land but the extinction of common and customary use rights on which many people depended for their livelihood" [Wood, 2002, p.108], and its early, most disruptive wave targeted pasture for sheep – Thomas More, "though himself an encloser, described the practice as 'sheep devouring men"' [Wood, 2002, p.108-109].

Let $\Xi(t) \in [0,1]$ denote the aggregate share of common/use-right land already enclosed, diffusing alongside improvement ideology – Wood ties the two processes together directly, since it is "reasons of 'improvement"' that were "cited systematically...as grounds for extinguishing traditional rights" [Wood, 2002, p.115] – with a discrete acceleration once state resistance to enclosure disappears:

$$\Xi(t+1) = \Xi(t) + \nu_1\big(1-\Xi(t)\big) + \nu_2\,\bar{\iota}(t)\big(1-\Xi(t)\big) + \nu_3 \mathbb{1}[t \geq t^*]\big(1-\Xi(t)\big), \tag{20}$$

where $\bar{\iota}(t)$ is the population-average of the ideology term $\iota_i(t)$ (§4.8) and t^* is a scenario parameter marking the point at which the state stops contesting enclosure at all: "in its earlier phases, the practice was to some degree resisted by the monarchical state...but once the landed classes had succeeded in shaping the state to their own changing requirements – a success more or less finally consolidated in 1688...there was no further state interference" [Wood, 2002, p.109], the same alignment of state and landlord interest already used to set θ (§4.6) – Brenner independently ties enclosure to the identical alliance, describing the English gentry as the very group "most concerned to undermine peasant property in the interests of enclosure and consolidation" [Brenner, 1976, p.71-72].

(HOW DO TENANTS ALL BENEFIT FROM COMMONS, THERE IS MENTION LATER IN PARAGRAPH THAT TENANTS COULD ENCLOSE COMMONS BUT STILL ARE EXPOSED TO RISK THIS FEELS WROGN,

SHOULD BE THAT COMMONS USED BY ALL AND THAT LANDLORD TRIES TO ENCLOSE AND THEN RENT THE LAND OUT)Enclosure's effect is external to the tenancy whose rent status changes: it extends the turnover hazard (4) with a third term, raising it for *any* nearby smallholder – Customary tenant or marginal Freeholder alike – who depended on commons access to make ends meet, whether or not their own parcel is the one being converted. A fixed share ρ_{commons} of parcels in each estate is flagged commons-dependent at $t = 0$, chosen at random independently of $\bar{\phi}_k$, and carries $\mathbb{1}[k \text{ commons-dependent}] = 1$ for the life of the run – operationalising Wood's "various other kinds of use rights" (quoted above) as a fixed structural feature of the manor, not one that itself grows or is assigned endogenously,

$$\eta_j^{\text{crisis}}(t) = \eta_1 \max\bigl(0, \bar{y} - y_j(t)\bigr) + \eta_2\, \Xi_{i(j)}(t) \cdot \frac{\bigl|\{k \in \mathcal{P}_j(t) :\ k \text{ commons-dependent}\}\bigr|}{|\mathcal{P}_j(t)|}, \tag{21}$$

which nests (4) exactly when $\Xi(t) = 0$. A household is exposed to its own lord's enclosure $\Xi_{i(j)}(t)$, and in proportion to the share of its holding that is commons-dependent rather than through a bare indicator; under the national rule of §4.12 every estate carries the same value and this reduces to $\Xi(t)$ exactly, so the distinction bites only under the local rule. This is what lets the model represent enclosure as the specific driver of dispossession *without* an individual tenancy conversion – the historical "growing plague of vagabonds, those dispossessed 'masterless men' who wandered the countryside" [Wood, 2002, p.108-109] – as distinct from, though reinforcing, the rent-conversion and engrossment channels above.

The indicator in (21) is written per parcel, but the hazard is applied per *household*, and once engrossment lets one household hold several parcels the two are no longer the same thing. The implementation therefore uses the *share* of a household's holding that is commons-dependent in place of the indicator, which coincides with (21) for a single-parcel holding and generalises it for a larger one: a household that has engrossed four parcels of which one carried common right is a quarter exposed, not fully exposed. Reported here because it is a departure from the equation as written, and because it makes enclosure's bite fall off as holdings consolidate, which is a substantive consequence rather than a bookkeeping detail.

A national process with a local driver. As specified, $\Xi(t)$ is a single aggregate and $\bar{\iota}(t)$ a population average, so enclosure has a timing but no location. This is faithful to Wood, who describes enclosure as a national movement in its legal and political aspect, but it sits awkwardly with the mechanism it is tied to: the improvement ideology of §4.8 diffuses *locally*, through the awareness neighbourhood, and enclosure is then the only mechanism in the model that is national while its own driver is local.

The switch `enclosure_rule` therefore admits a variant in which each estate carries its own $\Xi_i(t)$, driven by the mean disposition of that lord and the lords within the awareness radius, with the reported aggregate the parcel-weighted mean. It is not the baseline, and the reason is worth stating rather than burying in a robustness table. England's enclosure geography was largely set by pre-existing field systems – the open-field Midlands were the enclosable land, much of the south-east and west long since enclosed – and the model has no field-system layer. A locally diffusing Ξ_i therefore inherits the geography of conversion, because the two share a driver. **The arm is consequently informative in one direction only.** Two fronts that coincide under it say almost nothing, being close to a construction; two fronts that *diverge* despite the shared driver are a genuine separability result, and one that favours Shaw-Taylor. Reported on those terms in §5, with the national rule as the baseline throughout.

4.13 Vacancy resolution and population continuity

The mechanisms above are all triggered "at a vacancy" without yet specifying, for every vacancy, in what order a landlord's options are exercised, nor where the next occupant of a re-let tenancy comes from. Both questions are answered by distinguishing two kinds of vacancy already implicit in (4): the baseline hazard η_0, representing ordinary generational succession (the death of a tenant with an heir), and the excess hazard $\eta_j^{\text{crisis}}(t)$ – together with rent-default eviction under any tenure state (§4.10), or a tenant's own departure by migration (§4.8) – representing genuine dispossession with no customary claimant left to fall back on. This is exactly the distinction Brenner draws between ordinary conveyance, at which the landlord's only historical lever was the entry fine [Brenner, 1976, p.62], and demographic-collapse vacancies that a landlord "could simply appropriate...and add...to his demesne" [Brenner, 1976, p.61] because no one remained to contest them.

Conveyance vacancies (probability η_0 at any occupied parcel, each period). The tenancy is not structurally vacant: an heir succeeds the outgoing tenant by default, inheriting $\mathcal{P}_j(t)$ and a fraction ξ_{inherit} of their wealth, with capital reset to the traditional baseline k^{trad} (§4.10) (SHOULD IMPROVEMENT NOT BE INHERITED? IM NOT SURE IF IT IS AT THE MOMENT). This is precisely the moment of "conveyance" at which the landlord may still act (§4.6): the engrossment check and conversion decision below run exactly as at a crisis vacancy, so an entry fine can still be levied on the heir even though the parcel never stood empty.

Since §4.14 gives the household an explicit mortality rate, η_0 must be read strictly as the hazard of *conveyance* – the generational turnover of the head at which the tenancy is regranted – and not as a death rate, or the two would double-count the same deaths and the population would drain at $\eta_0 + m_j(t)$. It is accordingly population-neutral: the heir inherits the household at its current size $n_j(t)$, and the deaths of individual members are handled entirely by (23). The one exception is the option of impartible inheritance (§4.14), under which conveyance passes the holding whole to a single heir and the remaining $n_j(t) - 1$ members leave as a new landless household.

Extinction vacancies. A household that loses its last member (§4.14) leaves its parcels with no claimant of any kind. These are resolved exactly as crisis vacancies below, with the single difference that no dispossessed household joins the landless pool – there is nobody left to dispossess. This is escheat rather than eviction, and it is the mechanism Brenner identifies in the aftermath of demographic collapse, when a landlord "could simply appropriate...and add...to his demesne" [Brenner, 1976, p.61] land that no surviving tenant could contest. Because it takes precedence, an extinct household is never also counted as evicted or as conveyed.

Crisis vacancies. No customary claimant remains, so parcel k genuinely falls to the landlord's disposal. Resolution proceeds in a fixed order:

1. *Engrossment check* (§4.11): draw engrossment with probability $q_i(t)$. If it succeeds, k is absorbed into j^*'s holding and resolution ends here – no new tenant is created.

2. *Conversion decision* (§4.6): if not engrossed, draw conversion with probability $p_i^{\text{convert}}(t)$.

3. *Freehold diversion* (§4.7): if conversion is resisted (the draw in step 2 fails), draw $p_j^{\text{Freehold}}(t)$; on success the new occupant enters directly as Freehold rather than Customary.

4. *New occupant.* If neither engrossed nor converted, the parcel is re-let at its unchanged nominal rent $r_j^C(0)$ – custom attaches to the land, not the person.

   Tenure ratchets. This step applies only where the parcel is *still* customary. Tenure is a property of the parcel and moves in one direction: a leasehold parcel whose tenant fails is re-let as leasehold, never returned to custom. The distinction is not a detail. Read the other way – with every unconverted vacancy reverting to customary tenure regardless of the parcel's history – the customary sector becomes an absorbing state that the model falls back into faster than conversion escapes it, and no transition can accumulate however favourable the conversion decision is. Historically the point is not in doubt: the extinction of customary right is the irreversible process Brenner and Wood are describing, and §4.12's account of use-rights being extinguished rather than suspended says as much. If converted (step 2) or diverted to Freehold (step 3), the new occupant is drawn preferentially from the Landless pool $\mathcal{L}(t)$ (§4.15), subject to an entry cost $\chi \cdot \hat{r}_i(t)$ – the same entry-fine object introduced qualitatively in §4.6, now given a magnitude, where $\hat{r}_i(t)$ is estate i's own competitive rent benchmark (§4.10), the same figure the new tenancy will itself be charged if Leasehold. A Landless agent takes the vacancy if $w_\ell(t) \geq \chi\, \hat{r}_i(t)$, entering with $w_j(0) = w_\ell(t) - \chi\, \hat{r}_i(t)$, $k_j(0) = k^{\text{trad}}$, and $\iota_j^T(0) = 0$.

If step 4 requires a Landless entrant and none clears the entry cost, k joins a vacant-parcel queue, re-attempting steps 1–4 (with $\Delta_i(t)$ and $q_i(t)$ recomputed) every subsequent period until filled; a parcel in this queue produces no output and no rent, itself contributing to $\Delta_i(t)$. Land held by neither a tenant nor the queue does not otherwise exist in the model: every parcel is, at all times, either held, being engrossed, or in the queue.

This closes the model's population accounting without an unmodelled reservoir of prospective tenants: the customary sector reproduces itself through conveyance, matching the historical persistence of the customary peasantry across the period Brenner and Wood describe, while the Leasehold/Freehold sector's growth is bounded by the size of the Landless pool it itself generates through eviction and partition (§4.14, §4.10) – the "reserve army" channel of §4.15 is therefore also, endogenously, the channel that staffs new capitalist tenancies, not merely the wage-labour market. Every occupant of a re-let parcel comes from the Landless pool, so the model never conjures a tenant from nothing, and the cumulative exit share of §4.17(vi) remains a share of a determinate population.

4.14 Household demography: fertility, mortality and partition

Population in this period was not fixed, and an account of agrarian change has to say something about it. The difficulty is that the obvious way to model it is the one Brenner's 1976 intervention was written to reject: an aggregate law in which population grows with the food supply and presses back on the land, so that the outcome of agrarian history is read off a ratio of people to acres. A model in which such a law silently drove the transition would beg the question the paper is asking.

The resolution is to locate the demographic rates at the level at which Brenner locates causation – the individual household's position within a structure of property relations – and let the aggregate rate be whatever the composition of classes makes it. Nothing in what follows references a total population, a food supply, or a carrying capacity. What each household reads is its own per-head surplus relative to its own subsistence requirement,

$$g_j(t) = \begin{cases} \dfrac{y_j(t) - \rho_j(t) - \omega(t)\ell_j^{\text{hired}}(t) - n_j(t)c^{\text{head}}}{n_j(t)\, c^{\text{head}}} & \sigma_j(t) \in \{\text{Customary}, \text{Leasehold}, \text{Freehold}\}, \\[2ex] \dfrac{\omega(t)\bar{\ell} - c_{\ell,j}}{c_{\ell,j}} & \sigma_j(t) = \text{Landless}, \end{cases} \tag{22}$$

a dimensionless quantity, zero when the household exactly meets subsistence. (The arbitrary fine $F_j(t)$ of §4.10 is deliberately excluded, since it is itself levied on customary income above subsistence and would otherwise be charged twice.) The two branches are the substantive point rather than a technicality. A landholding household's demographic prospects are set by its land, its capital and the rent it owes; a landless household's are set by the wage. It follows that the reproduction rate of the population as a whole is determined by the distribution of property – the same land supporting a growing or a shrinking population according to how its surplus is divided – which is Brenner's claim about demography rather than Postan's, and which the model can now express instead of assuming away.

Fertility and mortality. Each member of household j independently gives birth with probability $b_j(t)$ (CLAUDE INVENTED PEASANT MITOSIS) and dies with probability $m_j(t)$,

$$b_j(t) = \min\big(b^{\max},\ m_0 + \beta^b g_j(t)\big), \qquad m_j(t) = m_0 + m_1 \max\big(0,\ -g_j(t)\big), \tag{23}$$

so that fertility is anchored at the baseline mortality m_0 where surplus is zero. A household exactly at subsistence is therefore stationary by construction; prosperity raises fertility to a ceiling $b^{\max}$; and a household in deficit is losing members to *both* sub-replacement fertility and excess mortality, the nutritional channel that m_1 scales. Births and deaths are drawn as Binomial$(n_j(t), b_j(t))$ and Binomial$(n_j(t), m_j(t))$, giving

$$n_j(t+1) = \min\big(n^{\max},\ n_j(t) - \text{deaths} + \text{births}\big), \tag{24}$$

with $n^{\max}$ a numerical guard. A household reaching $n_j = 0$ is extinct: if it holds land, its parcels become the extinction vacancies of §4.13.

Partition: how a household sheds members into the proletariat. A household with more hands than its land can use sends one out to work for wages. The member leaves when they are better placed outside the household than in it, both sides reckoned net of what it costs to keep them: inside, they add the marginal product of labour and eat per-head subsistence; outside, they earn the wage and must meet their own landless requirement. For the production function (18) this gives the condition

$$\underbrace{(1-\mu-\gamma)\frac{y_j(t)}{h_j(t)} - c^{\text{head}}}_{\text{net contribution inside}} \;<\; \underbrace{\omega(t)\bar{\ell} - c_{\ell,j}}_{\text{net position outside}} \;, \tag{25}$$

and a household satisfying it (or in per-head deficit, $g_j(t) < 0$) sheds one member with probability p^{part} per period, the new landless household taking a dowry w^{dowry} from its parent's wealth. Partition applies only to households holding land; a landless household splitting in two would change nothing, since both halves already live by the wage.

Three features of (25) are worth drawing out. It is a comparison the household could actually make from quantities it observes, rather than an accounting threshold imposed on it. It runs the classic peasant response in the right direction: a falling wage makes the right-hand side smaller, so families *retain* members and work the holding harder, while a wage that industry has bid up releases them – the self-exploitation of the family farm and its dissolution being the same mechanism read at two wage levels [Chayanov, 1966]. And because $y_j(t)$ depends on the household's own $\Phi_j(t)$, the productivity of a household's particular land is what sets how many people it can hold: good or consolidated land absorbs population, poor or fragmented land expels it, parcel by parcel and with no aggregate ceiling anywhere in the specification. This is also the channel through which the engrossment of §4.11 feeds back into the supply of wage labour, rather than merely coinciding with it.

4.15 The wage-labour pool: hiring, wages, and exit

A landless agent is not a sink. Brenner's consolidation mechanism only works because the larger holdings it produces need more hands to farm than a single household supplies: demesne land was "leased out to larger tenants...[who] in turn sub-let to smaller farmers or employed wage labourers" [Brenner, 1976, p.61]. The model must therefore specify what the landless do next, not merely how tenants come to lose their land.

Labour demand. A Leasehold or Freehold household hires up to the point at which the marginal product of labour equals the going wage, net of the labour its own members already supply, which for the production function (18) gives (WHERE THE HELL DOES THIS COME FROM)

$$\ell_j^{\text{demand}}(t) = \max\Big(0,\ \Big[\tfrac{(1-\mu-\gamma)\,\Phi_j(t)^{\mu} k_j(t)^{\gamma}}{\omega(t)}\Big]^{\frac{1}{\mu+\gamma}} - n_j(t)\,\bar{\ell}\Big), \tag{26}$$

so that it is precisely the successful, improving, consolidating tenants – Wood's "success would breed success" [Wood, 2002, p.100-101] – who hire, and unsuccessful ones who are hired. Family and hired labour are thus substitutes at the margin, and because only the hired part is charged at $\omega(t)$ in (13), they are not substitutes at equal cost: a large household working its own holding undercuts an equally large holding staffed by wage labour, which is the family farm's advantage and the reason (25) rather than a wealth threshold governs when it gives a member up. Demand has to reference $\omega(t)$: a rule in capital alone leaves nothing in the model responding to the price of labour, so the tâtonnement below has no equilibrium to find and the wage simply diverges. Correspondingly, the hiring tenant pays the wage bill $\omega(t)\ell_j^{\text{hired}}(t)$, which enters (13) alongside rent and subsistence; without it hired labour is free, and capital, output and wealth compound on one another without limit. Demand is a claim on a finite pool, not a guarantee: if aggregate demand $D(t) = \sum_j \ell_j^{\text{demand}}(t)$ exceeds the pool, hiring is rationed pro rata,

$$\ell_j^{\text{hired}}(t) = \ell_j^{\text{demand}}(t) \cdot \min\Big(1,\ \frac{S(t)}{D(t)}\Big), \qquad S(t) = \sum_{\ell \in \mathcal{L}(t)} n_\ell(t)\,\bar{\ell}, \tag{27}$$

so that $h_j(t) = n_j(t)\bar{\ell} + \ell_j^{\text{hired}}(t)$ in (18) never draws on more bodies than actually exist. Note that the supply $S(t)$ counts *bodies*, not households: a landless household of four supplies four labour endowments even though it bids once at an auction (§4.9). The unmet portion of demand – rather than a phantom labour supply – is exactly what the wage adjustment below prices.

Wage determination. The same excess demand $D(t) - S(t)$ that rationing absorbs in quantity is what moves price, through a simple tâtonnement,

$$\omega(t+1) = \omega(t)\Big(1 + \kappa\,\frac{D(t) - S(t)}{S(t)}\Big), \tag{28}$$

so that a labour supply growing faster than tenant demand for hired labour depresses the wage – the model's "reserve army" channel, consistent with the historical pattern of rents rising while wages stayed comparatively flat over the same period [Kerridge, 1953, Clark, 2002]. Because $S(t)$ is fed by the partition rule of §4.14 as well as by eviction, the wage is where the two halves of the theory meet: dispossession and household formation both price labour, and the wage in turn feeds back into (25) to govern how readily the next household gives a member up.

Landless income and exit. A landless household sells the labour endowment of every member at the going wage and feeds every member out of the proceeds,

$$w_j(t+1) = w_j(t) + n_j(t)\big(\omega(t)\,\bar{\ell} - c_{\ell,j}\big), \tag{29}$$

so that its per-head position, and hence its fate, is independent of its size. The requirement $c_{\ell,j}$ is drawn per household from $\mathcal{U}\big(c_\ell(1-\varepsilon_c),\, c_\ell(1+\varepsilon_c)\big)$ and fixed for its lifetime. This matters more than it appears to. With a single shared requirement, and a wage that is by construction global, every landless household in the model faces an identical net income, crosses zero on the same tick, and – under a deterministic counter – leaves on the same tick as well; the cohort behaviour described in §4.14 is the consequence. Departure is correspondingly a hazard in the *depth* of a household's deficit rather than a threshold in its duration,

$$p_j^{\text{exit}}(t) = 1 - \exp\Big(-\lambda^{\text{exit}}\,\frac{\max(0, -w_j(t))}{n_j(t)\,c_{\ell,j}}\Big), \tag{30}$$

measured in periods of the household's own consumption, so that a household barely under water rarely leaves and one deeply in debt almost surely does, and two otherwise identical households leave at different times. Exit is not a modelling convenience but the mechanism Brenner names directly: "agricultural improvement not only made it possible for an ever greater proportion of the population to leave the land to enter industry" [Brenner, 1976, p.67]. In the reverse direction, a landless household whose wealth clears the entry cost $\chi \cdot \hat{r}_i(t)$ of an open vacancy (§4.13) may re-enter directly as a Leasehold tenant – or, at a conveyance vacancy left without an heir, as a Customary one – giving the model a (theoretically vanishing, as land concentrates) channel of upward mobility rather than assuming the triad, once formed, is a closed caste system. The older deterministic rule, exit once $w_j(t) < 0$ for τ' consecutive periods with $\varepsilon_c = 0$, is retained as a switch for comparison.

The urban sector is a place, not a sink. Leaving for industry is a move, and treating it as terminal is not the neutral choice it appears to be. An absorbing urban state builds a one-way ratchet into the population accounting: the countryside can then only lose people to it, whatever its own vital rates, and the resulting decline is reported as though it were a finding.

An urban household therefore persists with its size and wealth, earns ω^U per member, pays c^{urban} per member, and is subject to the same vital rates (23) as everyone else on the per-head surplus those two imply. It returns to the landless pool with probability

$$p_j^{\text{return}}(t) = p^{\text{ret}} \cdot \min\Big(1,\ \max\Big(0,\ \frac{\big(\omega(t)\bar{\ell} - c_{\ell,j}\big) - \big(\omega^U - c^{\text{urban}}\big)}{c_{\ell,j}}\Big)\Big), \tag{31}$$

the same net-position comparison that governs partition (25), read in the opposite direction. It is scaled by the *size* of the advantage rather than triggered by its sign, because returning households enlarge the labour pool and so depress the very wage that drew them: on a bare sign test that feedback produces a churn loop, cycling people between town and country on a vanishing differential, rather than a response to conditions.

Two calibration points follow, and both are consequential enough to state rather than bury. ω^U is exogenous – the model has no industrial sector to derive it from, and inventing one would extend the paper well past the agrarian transition it is testing – so holding it fixed while $\omega(t)$ moves endogenously is what makes the direction of migration respond to conditions in the *countryside*, which is the causal direction the theory is about. And it is set equal to c^{urban}, making the urban per-head surplus exactly zero and, by the anchoring of (23) at m_0, the urban sector demographically stationary. That is the neutrality the channel is for: the towns neither breed nor consume a population of their own, so every movement in the rural/urban split is migration answering to rural conditions. Departing from equality is a substantive claim about urban demography that this model has no evidence for, and it is not a small one – at $\omega^U = 0.9$ against $c^{\text{urban}} = 1$ the towns lose about 1.9% of their people per period, which drained 74% of the *total* population over 200 periods and put a third of all parcels back into the vacancy queue.

The domestic market. Nor do urban households only supply labour: they are the market the remaining farms sell into. Wood's claim is that agrarian productivity produced "an increasing propertyless mass that would constitute both a large wage-labour force *and* a domestic market for cheap consumer goods" [Wood, 2002, p.103], and without the second half the exit channel is a pure drain in which the countryside never feels the consequences of its own dispossession. Writing $U(t)$ for the number of persons currently in the urban sector – a stock, since (31) lets them leave it – the produce price adjusts to the excess of their demand over what the market-exposed sector actually markets,

$$P(t+1) = P(t)\Big(1 + \kappa_P \, \frac{U(t)\, c^{\text{urban}} - Y^M(t)}{Y^M(t)}\Big), \qquad Y^M(t) = \sum_{j:\, \sigma_j \in \{\text{L,F}\}} y_j(t), \tag{32}$$

and market-exposed output is valued at $P(t)$ in (13) and (22). Customary production is not: it is largely eaten at home, so it is not revalued by a market the household does not sell into – which is itself a difference in kind between the tenures, and one that widens as $U(t)$ grows.

The point of specifying this rather than leaving "wage labourer" as a terminal label is that the theory's own account of *why* the transition matters beyond agriculture runs through exactly this population: rising agricultural labour-productivity is what "created a highly productive agriculture capable of sustaining a large population not engaged in agricultural production, but also an increasing propertyless mass that would constitute both a large wage-labour force and a domestic market for cheap consumer goods – a type of market with no historical precedent. This is the background to the formation of English industrial capitalism" [Wood, 2002, p.103]. The exit channel above is what lets the model, in principle, speak to that claim rather than only to the agrarian triad in isolation.

4.16 Parameters and initial conditions

Table 2 collects every free quantity introduced above with a placeholder starting value. Because §4.17 treats the model's outputs as a qualitative plausibility check rather than a fitted calibration target, these are deliberately round starting points for the first runs and the sensitivity sweeps that follow, not estimates drawn from data.

Symbol	Value	Meaning
Space and population (§4.4)		
L	155	lattice resolution: L cells span the longer axis of England's bounding box and the shorter axis takes as many square cells of the same size as it holds, giving a 155×136 grid of which 7,417 cells survive the boundary mask
N_L	10 per county	landlords/estates, seeded within each historic county so that no estate straddles a county boundary; 39 counties give 390 estates
r	5	landlord awareness radius, in lattice cells
ζ	2	within-county fertility jitter (§4.4)
Initial agent state (§4.3)		
$W_i(0)$	100	landlord initial wealth
C_i	$\mathcal{U}(0.85, 1.05)\times$ rent roll	consumption requirement, scaled to the estate (§4.6)
initial σ_j shares	0.95 / 0.05 / 0 / 0	Customary / Freehold / Leasehold / Landless
$w_j(0)$	10	tenant initial wealth
k^{trad}	1	traditional (non-market) capital baseline
$r^C_k(0)$	$\mathcal{U}(0.5, 1.5)$	customary rent at grant, drawn per *parcel*
$\bar{\ell}$	1	labour endowment *per household member*
$n_j(0)$	3	initial household size (§4.14)
μ_j	$\mathcal{U}(0, 0.3)$	tenant mobility propensity
Exogenous processes (§4.5)		
$\pi(t)$	0.02	inflation rate per period
ρ	0.1	fertility regrowth rate
$\bar{\phi}_k$	county baseline $\pm\zeta$ (§4.4)	carrying capacity, from ALC grades [Natural England, 2021]
$\epsilon_k(t)$	Bernoulli(0.05) $\times\, \mathcal{U}(0, 2)$	ecological shock
$\bar{y}$	2	subsistence output threshold
η_0, η_1, η_2	0.02, 0.05, 0.03	turnover hazard coefficients
Conversion, resistance and Freehold (§4.6, §4.7)		
$\alpha_0, \alpha_1, \alpha_2, \alpha_3$	-4, 2.5, 3, 1	conversion logit coefficients; pressure necessary, not sufficient
ϖ	0.05	hazard of reopening terms on a sitting tenant (§4.6)
θ	2 (England) / 6 (France)	state–landlord alliance regime
ξ_0, ξ_1	0.3, 0.5	ideology discount on resistance
$\lambda_0, \lambda_1, \lambda_2$	-5, 0.3, 1	Freehold-diversion logit coefficients
t^*	100	period at which state resistance ends
χ	2	entry cost, in periods of rent
ξ_{inherit}	0.5	wealth fraction retained at succession
Spread (§4.8)		
β_1, β_2	0.002, 0.15	ideology diffusion: autonomous, and from neighbour conversions
$\iota_i(0)$	0	initial ideology exposure
Tenant economy (§4.10)		
θ_{rent}	0.3	competitive-benchmark extraction share, $\hat{r}_i(t) = \theta_{\text{rent}}\hat{y}_i(t-1)$
β^T	0.1	necessity-driven growth rate of tenant improving disposition $\iota^T_j(t)$
ς	0.02	decay of the improving disposition absent competitive pressure
φ	0.5	arbitrary-fine rate on customary income above subsistence
ϱ_k	0.05	capital depreciation
$\iota^T_j(0)$	0	initial tenant improving disposition (all tenure states)
$\bar{s}$	0.3	reinvestment-rate ceiling, scaled by $\iota^T_j(t)$
$w^{\min}$	0	wealth floor below which no reinvestment occurs
c_j	1	tenant subsistence requirement, per household (single-body population rules only; the household rule charges $n_j(t)\, c^{\text{head}}$ instead)
μ, γ	0.3, 0.3	Cobb–Douglas fertility/capital exponents
τ	3	periods of default tolerated before eviction
Engrossment and enclosure (§4.11, §4.12)		
ψ_0, ψ_1, ψ_2	-2, 3, 0.1	engrossment logit coefficients
ρ_{commons}	0.2	share of parcels flagged commons-dependent
ν_1, ν_2, ν_3	0.01, 0.05, 0.1	enclosure diffusion coefficients
$\Xi(0)$	0	initial enclosure share
Labour market and the domestic market (§4.15)		
κ	0.1	wage adjustment speed
$\omega(0)$	1	initial wage
c_ℓ	0.8	mean landless subsistence requirement, per head
ε_c	0.25	half-width of the per-household draw on $c_{\ell,j}$
λ^{exit}	0.35	exit-hazard scale in (30)
τ'	3	periods of negative wealth before exit (deterministic rule only)
c^{urban}	1	food demanded, and subsistence owed, per urban head
ω^U	1	urban wage per head; set to c^{urban} so the towns are

Two of the demographic values are exceptions to the "round starting point" rule above, in that the model is qualitatively wrong at the obvious choice, and it is worth saying why rather than leaving them to look arbitrary.

$n_j(0) = 3$ rather than 1. Once mortality applies per member, a household of one is extinguished by a single death: at $m_0 = 0.02$ that is a 91% chance of the line failing within 120 periods. Runs started from single bodies lose roughly half their households to extinction before the transition is under way, and the parcels those households held cannot be re-let because the labour pool is not large enough to supply takers – reproducing, by a different route, the vacancy pathology that §4.14 records for the closed population. A household line is not as fragile as a person, and starting from an established family is what makes that distinction operative.

$c^{\text{head}} = 0.5$ rather than the single-body subsistence $c_j = 1$. This constant fixes where partition bites, and therefore how much of the model's proletarianisation is demographic at all. Output rises with labour as $n_j^{1-\mu-\gamma}$ while subsistence rises as n_j, so a household on one parcel becomes unviable at roughly $n_j = (\bar{y}^{\text{parcel}}/c^{\text{head}})^{1/(\mu+\gamma)}$; against a mean output near 1.8 per parcel, 0.5 places that threshold at four to five members, giving mean household sizes of two to three, an extinction rate of a fraction of a percent per period, and mild aggregate population growth. Raising it toward 0.7 drives households back toward single bodies and multiplies extinctions – the failure mode the household unit exists to avoid; lowering it toward 0.45 produces markedly faster growth. It is reported here as a calibrated quantity, and belongs in the sensitivity analysis accordingly.

4.17 Emergent outputs and validation targets

Because the model is a validity test rather than an illustration, its outputs are defined as things that could fail to match the theory's claims. At minimum: (i) the share of tenancies in each tenure state over time, now including Freehold alongside Customary, Leasehold and Landless; (ii) *farm*-size concentration – the Gini coefficient of tenant holding size $A_j(t) = |\mathcal{P}_j(t)|$, which the engrossment mechanism of §4.11 predicts should rise over the run – kept *explicitly distinct* from landlord-estate-size concentration, which §4.11 treats as a fixed initial condition rather than an emergent target, so the model is not credited with reproducing a claim the theory itself does not make; (iii) whether conversion spreads with a distance-dependent lag or occurs simultaneously – the direct test of the "localised seed" claim, measured as the semivariogram of first-conversion time over the lattice, which asks whether parcels further apart convert further apart in time without requiring any point to be nominated as the origin of the transition; (iv) aggregate output $Y(t) = \sum_j y_j(t)$, compared *qualitatively* against the historical rent, wage and grain-output series assembled from secondary sources [Clark, 1991, Clark, 2002, Kerridge, 1953], used as a plausibility check on the shape of the transition rather than as a calibration target; (v) a direct comparative test of Brenner's own argument, by re-running the model with θ set to a "France" regime (institutional resistance systematically prevails), checking not only whether the Leasehold triad fails to emerge but whether the same displaced pressure instead resolves into entrenched Freehold smallholding (§4.7), as the theory predicts: "in England, it was precisely the absence of such rights that facilitated the onset of real economic development" [Brenner, 1976, p.74-75]; (vi) the composition of the landless pool over time – the share still seeking agricultural hire, the wage series $\omega(t)$ against the historical rent/wage ratio, and the cumulative share that has exited the rural model – as the model's operationalisation of the claim that agrarian improvement is what freed population "to leave the land to enter industry" [Brenner, 1976, p.67]; (vii) the enclosure share $\Xi(t)$ against the timing of dispossession implied by tenure-state shares, testing whether the model reproduces enclosure and rent-conversion as related but empirically separable drivers of landlessness, rather than a single conflated process; and (viii) the population as a *composition* – the share of all living persons who are Customary, Leasehold, Freehold, Landless or urban, against a total that is itself an outcome (§4.14). The composition is the target rather than the rural head count, for two reasons. Read alone, a falling rural population cannot distinguish a countryside losing people to towns from one failing to reproduce, and the theory makes a claim about the first and no claim about the second. And the tenure shares of (i) are shares of *tenancies*, so they are silent on how many people stand behind each one; a Leasehold sector that has consolidated onto larger farms worked by larger households is a different outcome from one that has not, and only a per-person denominator shows it. The qualitative pattern the theory implies is a rising total, a rising urban share, and a rural share whose *internal* composition shifts from Customary toward Leasehold and Landless – and it is a real possibility that the model produces the tenure shift without the population shift, which would be a negative result about the sufficiency of the mechanisms rather than a technical failure.

4.18 The historical series used as plausibility checks

Targets (iv) and (vi) above are read against four secondary series, listed in Table 3 so that what is being compared against what is visible rather than buried in a citation. Kerridge covers the model's own window; the Clark series are longer and are used for the shape of the movement rather than for its level.

Source	Series	Period	What it is read against
Kerridge [Kerridge, 1953]	Rent	1540–1640	$\hat{r}(t)$ and the mean real customary rent; the only series covering the model's own window
Clark [Clark, 2002]	Land rental values	1500–1914	The same, over a longer run, for the shape of the movement
Clark [Clark, 1991]	Yields per acre	1250–1860	Aggregate output $Y(t)$ and output per parcel
Clark [Clark, 2004]	Prices and wages	1209–1914	The wage series $\omega(t)$, and the rent/wage ratio of target (vi)

Table 3: The secondary series behind validation targets (iv) and (vi). Read for direction and relative timing, not for level: §4.16 disclaims calibration, so a mismatch in level is not a failure and a mismatch in direction is.

Why these three quantities together rather than any one of them. Brenner's landlord–tenant symbiosis predicts that rents and output rise *jointly* – both classes accumulating – while wages lag both; continental squeezing is the pattern in which rent rises and output does not [Brenner, 1976, p.48, p.65]. The triple is therefore the empirical shape that separates the two cases, and no single series does. The extraction share θ_{rent} (§4.10) is the parameter that moves the model between the two cases, so the historical panel is the shape against which that setting is judged.

5 Results

Results are organised by research question rather than by figure, and every figure below is produced by one named scenario or comparison of scenarios from the ladder of §5.1. Two conventions hold throughout. First, every series is a mean across seeds with a 95% confidence interval on that mean, because every mechanism in the model is stochastic and a single run cannot distinguish a mechanism from a lucky draw; where the interval on the mean is narrow while individual runs diverge, the per-seed panel is reported alongside it, since a tight interval on a bimodal distribution is a statement about arithmetic rather than about the model. Second, an ablation is always reported against the same seeds as its baseline, so that a difference between arms is not a difference between draws.

Provenance and reading conventions. Everything below comes from a single suite, run at 64 seeds per arm (seeds 0–63) with the estate layout held fixed across seeds within an arm, so that a difference between arms is a difference in event history rather than in geography. Every quoted difference is a *paired* difference: seed s under the ablation minus seed s under the baseline, averaged over the 64 seeds, reported as $\Delta \pm$ the 95% half-width on that mean, and accompanied where it matters by an exact two-sided sign test that assumes nothing about the shape of the difference distribution. Outcomes are also classified per seed rather than only averaged: a seed is *transitioned* if its final Leasehold share reaches the completion criterion of a Leasehold majority (≥ 0.5), *failed* if that share stays below 0.2, and *partial* in between. The classification is reported because a mean of 0.5 can describe every run half-converted or half the runs converting fully and half not at all, and those are different claims about the model. No arm in the suite is bimodal in the sense that matters – mass at both extremes with little in between – so the means quoted here can be read as means.

Two qualifications are owed at the outset. The pairing convention bought less than expected: across every group the ratio of the unpaired to the paired interval sits near unity (median ≈ 1.0, range 0.89–1.43) rather than in the range of three to five that a large shared seed effect would produce. With the lattice already held fixed across seeds, the largest common component of variation has been removed before pairing can act on it, and the pairing is therefore a safeguard here rather than a source of precision. Separately, two of the suite's fifteen scenario groups did not complete: the two-dimensional RQ4 frontier (a 8×8 cross of the terms-reopening hazard against the arbitrary-fine rate) and the structural-robustness group that varies lattice resolution and estate granularity. Both are the heaviest groups in the suite and both terminated without writing output. Their absence is noted at the points where it bears on a conclusion, in §5.5 and §5.3 respectively.

5.1 The complexity ladder

The mechanisms of §4 interact, and a run with all of them switched on cannot say which is responsible for what. The scenarios of `model_scenarios/scenarios.yaml` therefore build the model up in tiers, each adding one mechanism to the one before, so that a dynamic can be attributed to the tier at which it first appears. Tier 0 is the bare exogenous skeleton – inflation eroding fixed customary rents against no conversion at all – and each subsequent tier restores one element: the conversion decision, then the spread channels, then engrossment and competitive allocation, then enclosure, then the labour market and the urban sector, then endogenous demography. Every tier runs the full multi-seed replication. The ladder is reported once, as Figure 2, and thereafter treated as the reference against which the question-specific arms are read.

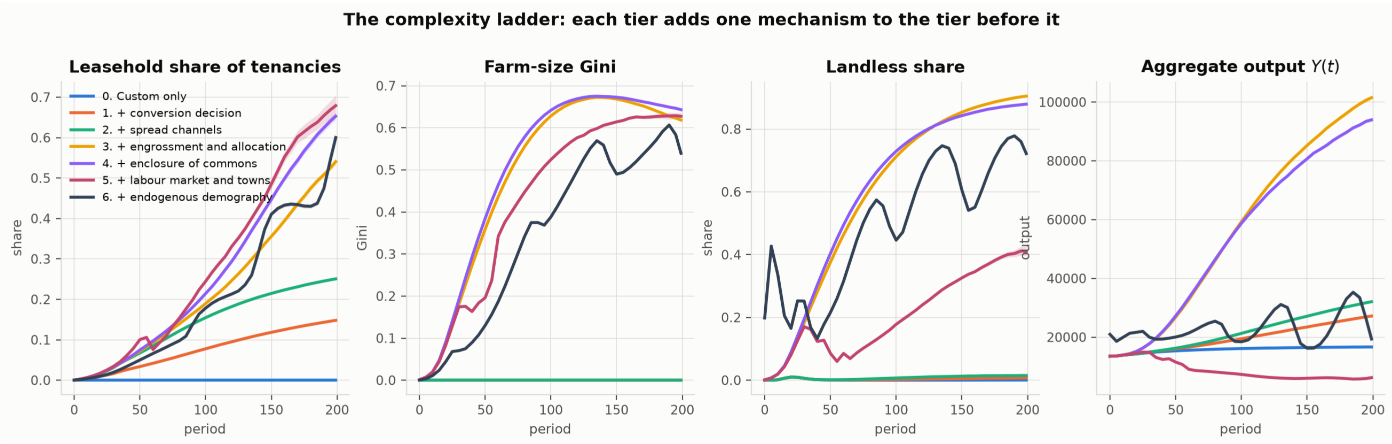

Figure 2: The complexity ladder. Each panel adds one mechanism to the tier before it; the series shown are the Leasehold share, the farm-size Gini, the landless share and aggregate output. A dynamic first appearing at tier n is attributable to the mechanism tier n introduces, which is what a single all-mechanisms run cannot establish.

Tier 0 behaves as the design requires: fiscal pressure rises and nothing happens, the Leasehold share staying at exactly zero in all 64 seeds, so no transition is being produced by anything other than the conversion decision. Restoring that decision (tier 1) moves the final Leasehold share to 0.148 ($\Delta = +0.148 \pm 0.001$), and restoring the two spread channels (tier 2) adds a further $+0.103 \pm 0.001$. Neither tier transitions a single seed: at tier 2 the model converts a quarter of tenancies and stops well short of a Leasehold majority in every replicate.

The decisive step is tier 3, which adds engrossment and competitive allocation together. It contributes $+0.290 \pm 0.008$ – more than twice any other tier – and it is where the regime changes, carrying the model from $0/64$ transitioned at tier 2 to $60/64$ at tier 3. It is also where the distributive outcomes appear at all. The farm-size Gini is exactly 0.000 through tier 2 and 0.619 at tier 3, and the landless share moves from 0.015 to 0.905 across the same step. Concentration and dispossession in this model are therefore not gradual accompaniments of conversion: they are switched on by a single mechanism, and conversion without it produces neither. Enclosure of the commons (tier 4) adds $+0.112 \pm 0.012$ and completes the transition in all 64 seeds.

Two later tiers qualify the picture. The labour market and urban sector (tier 5) do not change the tenure outcome at all – $+0.026 \pm 0.027$, sign test $p = 0.53$ – but they cut the landless share almost in half, from 0.879 to 0.411, by providing an exit that the earlier tiers denied. Endogenous demography (tier 6) is the only tier that moves population at all (the final-to-initial ratio rises from exactly 1.000 to 1.411), and it *reduces* the Leasehold share by 0.079 ± 0.025 while returning the landless share to 0.613. A growing population supplies takers for vacancies that would otherwise have gone to engrossment, which slows concentration and, through it, conversion.

5.2 RQ1: competitive tenancy or secure property

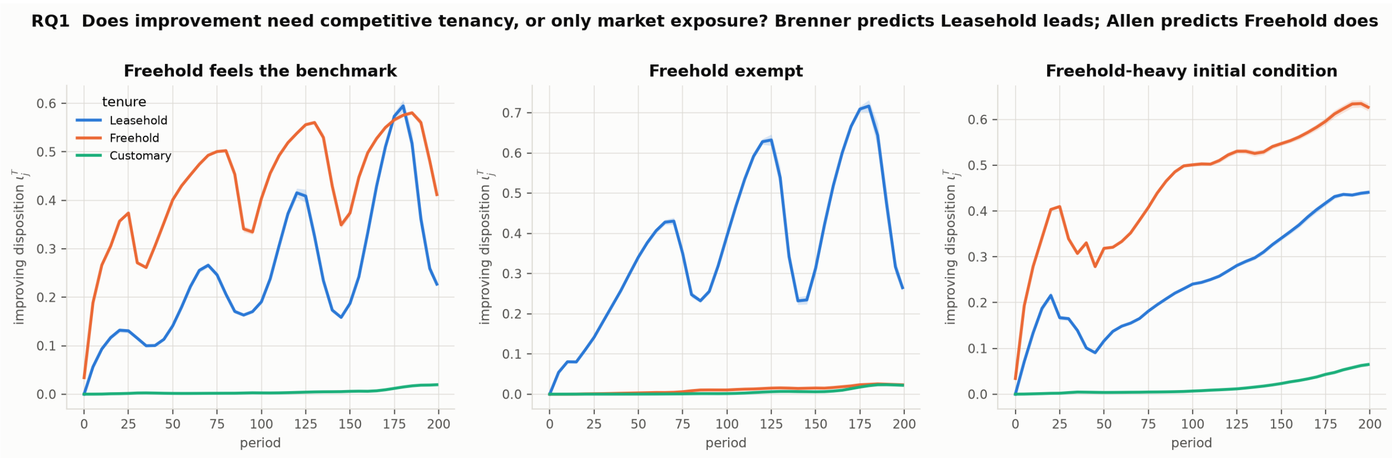

Figure 3: Improving disposition $\iota_j^T(t)$ by tenure state, Leasehold against Freehold on the same land in the same runs. Brenner's landlord–tenant symbiosis predicts Leasehold leads; Allen's yeoman predicts Freehold does. The middle panel repeats the comparison with `freehold_shadow_rent` disabled, which removes the competitive yardstick from Freehold entirely; the right-hand panel varies the initial tenure composition instead.

The comparison does not divide as either side of the debate frames it. Where Freehold feels the competitive benchmark (Figure 3, left), the Freehold improving disposition runs *above* the Leasehold one for the whole run, which is Allen's prediction rather than Brenner's. But where Freehold is exempted from that benchmark (centre), the Freehold series flattens onto the Customary one at around 0.02 while Leasehold rises to 0.71, which is Brenner's prediction rather than Allen's. The same reversal appears in accumulation (Figure 4): exposed Freeholders accumulate several times the capital of Leaseholders, because they face the yardstick without surrendering the surplus as rent.

What separates the two panels is not the security of the tenure but the presence of the yardstick, and the arm-level differences say the same thing. Exempting Freehold raises the aggregate transition by $+0.082 \pm 0.009$, because land that would have improved under the benchmark instead stagnates and is converted; whereas starting the run with a far larger Freehold sector changes the outcome not at all ($+0.005 \pm 0.017$, $p = 0.90$). The stock of secure property is immaterial; the exposure of that property to a competitive rent is what does the work. The model's answer to RQ1 is accordingly that the improvement dynamic requires market exposure and not competitive tenancy, which is a position intermediate between Brenner's and Allen's and one that the two panels of Figure 3 bracket rather than a single run could state.

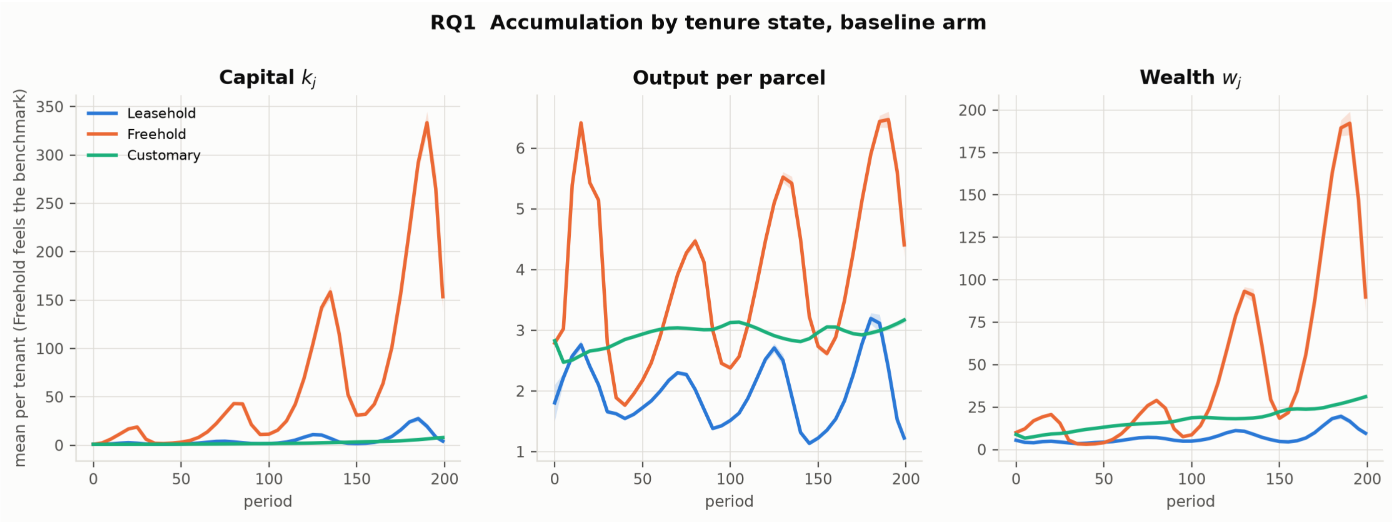

Figure 4: Capital, output per parcel and wealth per tenant by tenure state in the baseline arm, where Freehold feels the competitive benchmark. The Freehold series is the one that accumulates; the oscillation in all three is the ecological cycle of §4.5 propagating through investment.

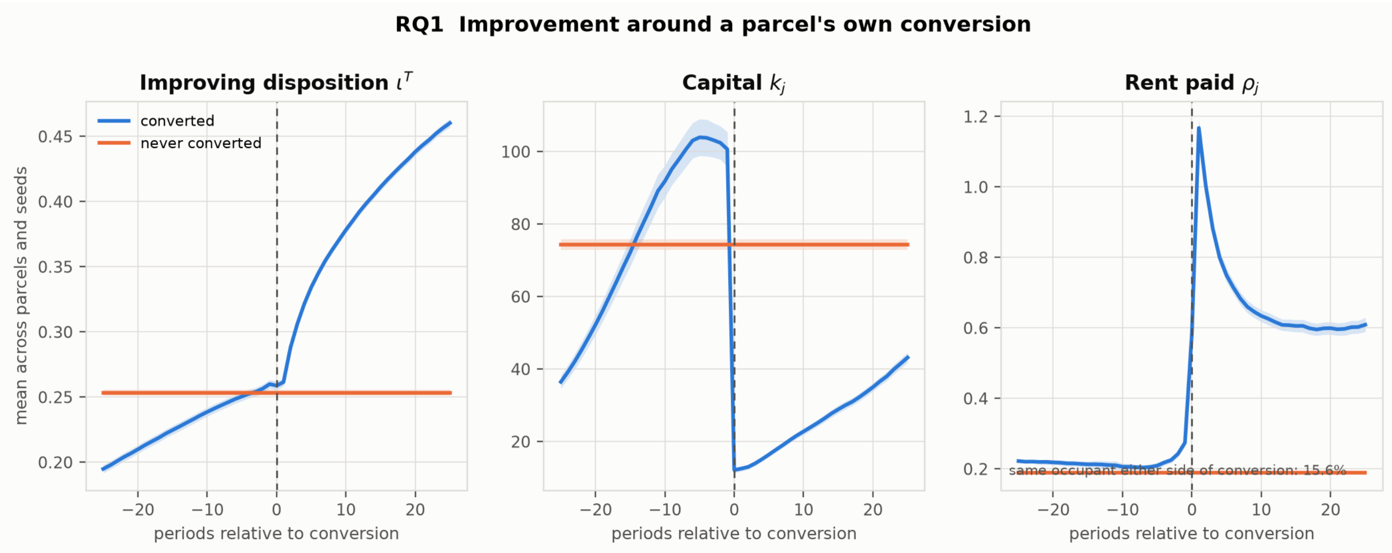

Figure 5: The within-tenure half of RQ1: every converted parcel aligned on its own conversion date, against parcels that never converted (flat reference). Reported with the occupant-continuity diagnostic, since a conversion that replaces the sitting tenant records a change of occupant rather than a change of disposition (§4.13).

Aligning each parcel on its own conversion date (Figure 5) shows the mechanism operating on the land rather than on the aggregate. Rent paid is flat at about 0.21 for twenty-five periods before conversion, spikes to 1.17 in the period after it, and settles at 0.61 – roughly triple the never-converted reference level of 0.19. The improving disposition of converted land rises from 0.195 twenty-five periods before conversion to 0.260 at the conversion date, crossing the never-converted level of 0.253 only at conversion, and then to 0.460 by twenty-five periods after. Capital behaves in the opposite direction at the event: it climbs to 104 in the five periods before conversion, collapses to 12 at the conversion date, and has recovered only to 43 twenty-five periods later, against a never-converted level of 74.

That collapse is compositional and must be read as such. The occupant-continuity diagnostic reports that the same household sits on the parcel either side of conversion in only 15.6% of cases; the remaining 84.4% of conversions install a new entrant who stocks the holding from scratch at k^{trad}. The post-conversion rise in the improving disposition is therefore substantially a statement about *who* now farms the parcel rather than about a sitting tenant's changed disposition, and the paper's RQ1, phrased about a tenant's own conversion, is answered by the model only in the weaker form: conversion raises the improving disposition of the land, principally by replacing its occupant. The pre-conversion capital peak is the complementary half of the same fact – parcels are selected into conversion after a period of accumulation, because accumulation is what raises the rent a neighbour will bid.

5.3 RQ2: spreading or simultaneous

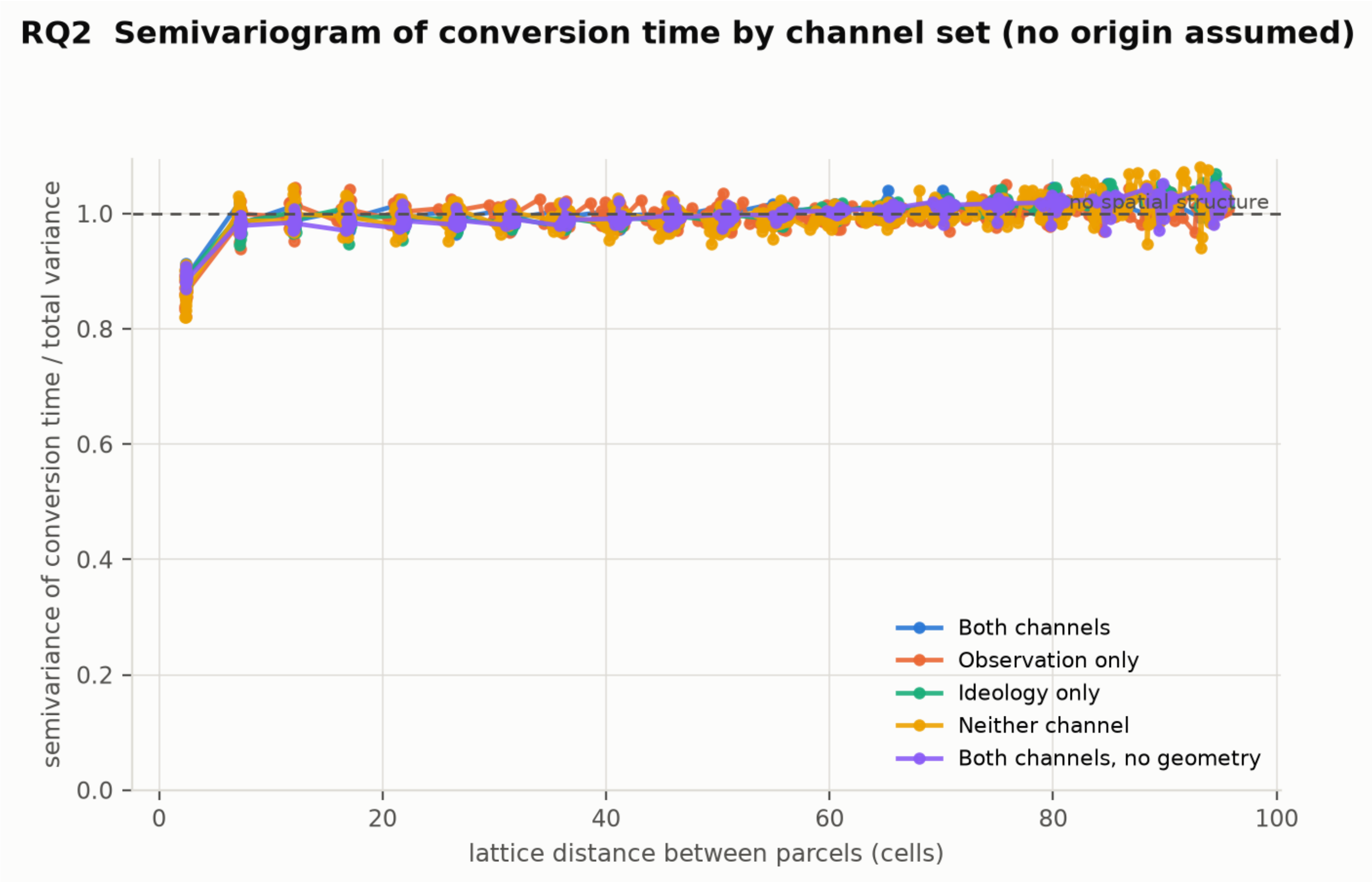

Figure 6: The semivariogram of first-conversion time: half the mean squared difference in conversion time between pairs of parcels, against the lattice distance separating them, normalised by the total variance. No origin is assumed. A curve rising from a low value at short distance is a spreading front, and the distance at which it flattens is the spatial scale of the process. A flat line at unity throughout is simultaneity: parcels one cell apart convert as differently as parcels a country apart.

This is the paper's clearest negative result, and Figure 6 states it without arithmetic. Under the full model the semivariogram leaves the origin at 0.89 and is indistinguishable from unity by the second distance bin; it does not rise from a low value, and there is no scale at which it flattens because it is flat almost immediately. The three summary numbers agree. The nugget share – semivariance at the shortest separation, over total variance – is 0.889 ± 0.004, meaning that neighbouring parcels differ in conversion time by nearly as much as parcels at opposite corners of England. The normalised slope is $+0.00072 \pm 0.00008$ variance shares per lattice cell, statistically distinguishable from zero but small enough that traversing the whole 155-cell lattice accounts for about a ninth of the variance. The range at which the curve reaches 95% of total variance is 7.4 cells. A high nugget with a near-zero slope and no spatial scale is precisely the configuration the measure was built to identify as simultaneity, and the model meets it.

The county-level film of Figure 7 shows the same thing in the form the question is usually asked in. Leasehold does not advance across England from anywhere; it deepens more or less everywhere at once, with the whole country lightening together between $t = 33$ and $t = 199$ and the modest regional contrast that does emerge by $t = 199$ arriving late and not travelling.

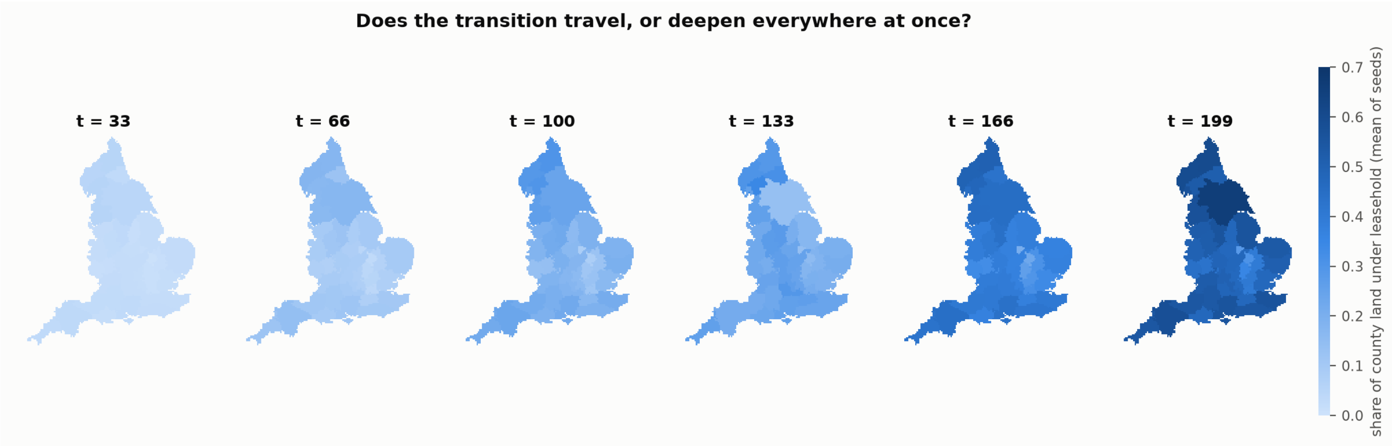

Figure 7: Share of county land under Leasehold, averaged over seeds, at six points through the run. If the transition travelled, the sequence would show a front; it shows uniform deepening instead.

Decomposing the two transmission channels sharpens rather than rescues the result. Removing the ideological channel and retaining observation changes the outcome not at all ($+0.001 \pm 0.008$, $p = 0.26$); removing observation and retaining ideology costs -0.181 ± 0.009; removing both costs -0.178 ± 0.009, which is statistically the same arm as removing observation alone. The observation channel carries the entire transmissible effect and the ideological channel carries none of it. What the observation channel supplies, however, is not spatial propagation but a level shift: its removal lowers the nugget share from 0.889 to 0.865, so the field becomes marginally *more* locally coherent when transmission is switched off, not less. Figure 8 makes the point spatially: the paired difference between the no-channel arm and the baseline is a nearly uniform -0.19 across the whole of England, with no region distinguishing itself. The channels change how much conversion happens, not where or in what order.

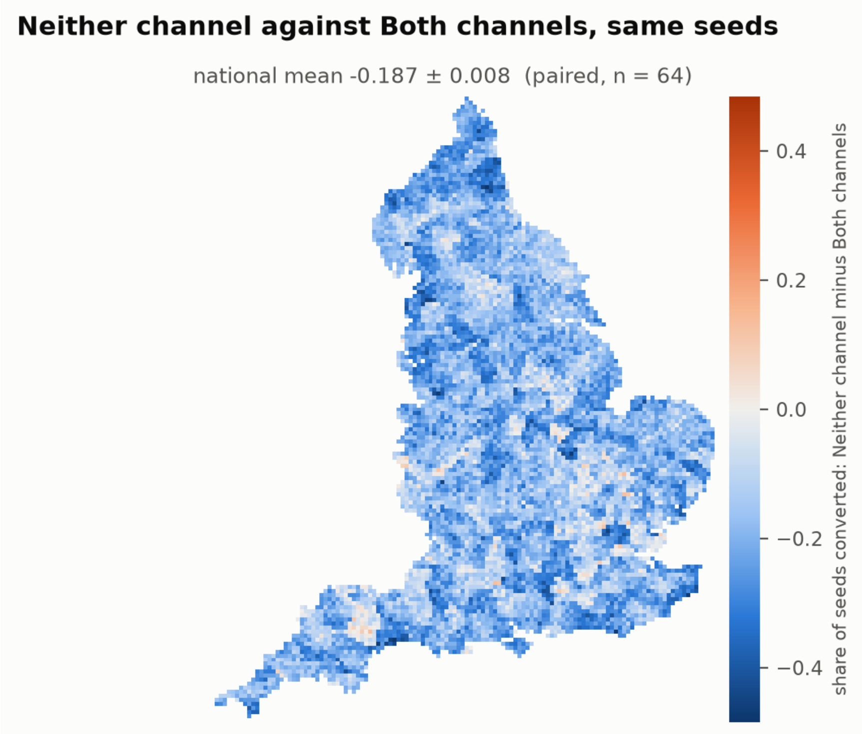

Figure 8: Both transmission channels disabled, differenced against the baseline parcel by parcel on matching seeds. A transmission mechanism with spatial content would leave a spatially structured difference; this one leaves a uniform level shift.

Two controls bear on whether this is a finding or an artefact of measurement. Destroying the geometry of the awareness graph while leaving every lord's number of contacts intact changes the outcome by $+0.005 \pm 0.007$ ($p = 0.17$) and the nugget by -0.0004: if spatial transmission were producing the result, randomising who watches whom would have to disturb it, and it does not. That arm was designed to demonstrate that the semivariogram *would* have detected a front had there been one, and it cannot do so here, because no arm in the suite produces a front for it to flatten. The instrument's validation therefore rests on the synthetic fields of `tests/test_model.py`, where a constructed radial front returns a nugget of 0.006 and a slope of $+0.090$ and a shuffle of the same conversion times returns a nugget of 1.006 and a slope of -0.001. That is an adequate validation, but it is a unit test rather than a scenario arm, and the stronger in-suite demonstration is not available. The second control is the one that did not run: the structural group varying lattice resolution and estate granularity would have established that a range of 7.4 cells is a fact about the process rather than about a 155-cell lattice with an awareness radius of 5.0. That the measured range sits at roughly 1.5 times the awareness radius is suggestive precisely because it is the quantity that group was built to test, and the result should be read as provisional until it has been.

Subject to that, the answer to RQ2 is that Brenner's stated mechanisms generate a simultaneous transition and not a spreading one. The account requires an inter-estate channel of a kind it does not supply, and the model's own transmission channels do not stand in for one: they raise the level of conversion without giving it spatial order.

5.4 RQ3: necessary, sufficient or permissive

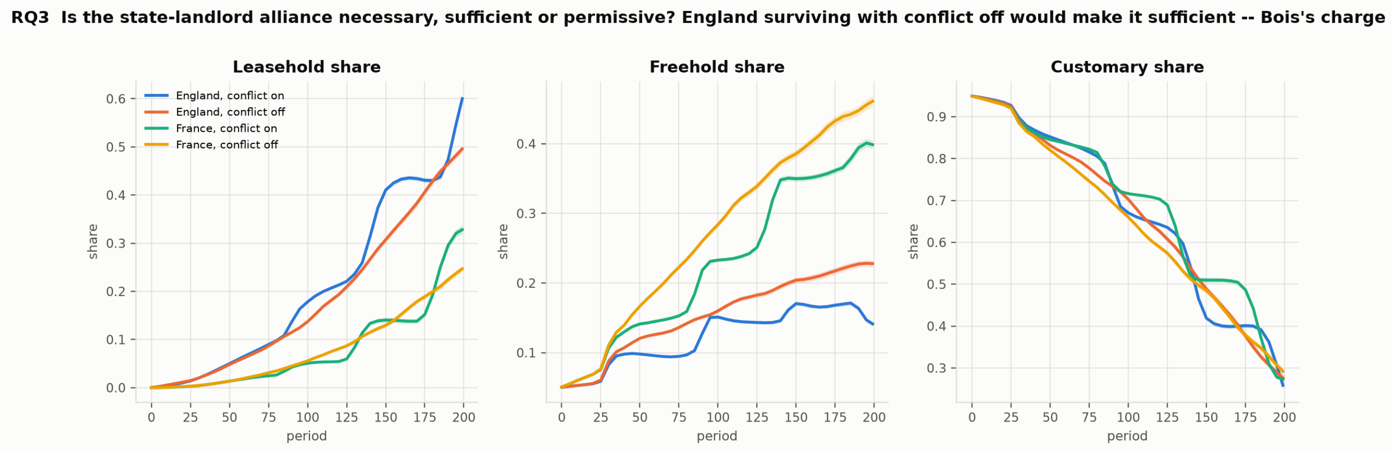

Figure 9: The 2×2 of RQ3: England ($\theta = 2$) and France ($\theta = 6$) crossed with the class-conflict mechanism enabled and disabled ($\alpha_1 = 0$, $\varphi = 0$). The England outcome surviving the disabled column would make the political condition sufficient on its own, which is Bois's charge; θ merely rescaling a transition that occurs in both columns would make it permissive.

The class-conflict mechanism is neither necessary nor sufficient in this model. It is not sufficient, because France with conflict fully enabled transitions in 0 of 64 seeds and ends at a mean Leasehold share of 0.328; institutional resistance at $\theta = 6$ suppresses the transition whatever the intensity of conflict. It is not necessary, because England with conflict disabled still transitions in 29 of 64 seeds and ends at 0.496, within a whisker of the completion criterion. What the mechanism does is move the threshold: England loses 0.105 ± 0.009 when conflict is disabled, France loses 0.272 ± 0.011 relative to England when resistance is raised, and an arm with both changes loses 0.353 ± 0.009 – slightly less than the 0.377 that strict additivity would predict, so the two conditions substitute weakly rather than reinforcing one another.

That headline understates the class-conflict channel, and the reason is worth stating because it is a property of the operationalisation rather than of the model. Disabling conflict here means setting $\alpha_1 = 0$ *and* $\varphi = 0$ together, and RQ4 shows that those two changes pull in opposite directions: removing the arbitrary fine alone raises the final Leasehold share from 0.600 to 0.798, while removing the fiscal-pressure term alone lowers it from 0.600 to 0.400. The composite ablation therefore cancels most of its own effect, and the -0.105 reported above is the residue. Read on the single-parameter margin instead, the fiscal-pressure channel is worth about -0.200, roughly twice what the 2×2 attributes to the whole conflict mechanism. The 2×2 is the right design for the question Bois poses, but its "conflict off" column should not be read as isolating class conflict.

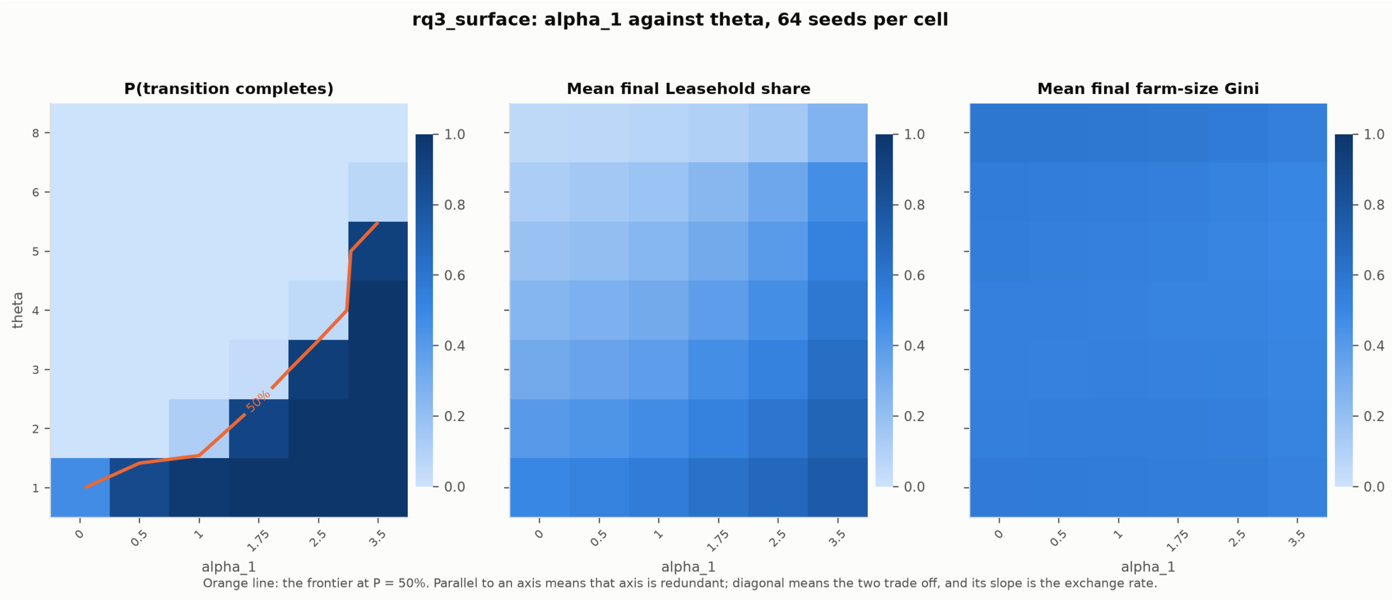

Figure 10: Conflict intensity α_1 against institutional resistance θ, 64 seeds in every cell. The left panel is the probability that the transition completes, with the frontier at $P = 0.5$ drawn on it; a frontier parallel to an axis would make that axis redundant, and a diagonal one means the two conditions trade off at the slope shown.

The surface of Figure 10 states the trade-off directly, and it is a clean diagonal over most of its range: the completion frontier runs from $\alpha_1 \approx 0.1$ at $\theta = 1$ to $\alpha_1 \approx 3.4$ at $\theta = 5.5$, so through the middle of the parameter space each additional unit of institutional resistance can be offset by roughly 0.7 additional units of conflict intensity. Neither axis is redundant. But the exchange is not indefinite: at $\theta = 8$ no attainable conflict intensity completes the transition, $P = 0$ even at $\alpha_1 = 3.5$, and the frontier leaves the top of the parameter space rather than continuing. The state–landlord alliance is therefore permissive over a wide range and necessary beyond it, which is a more specific claim than any of the three the question offers.

5.5 RQ4: how secure custom could have been

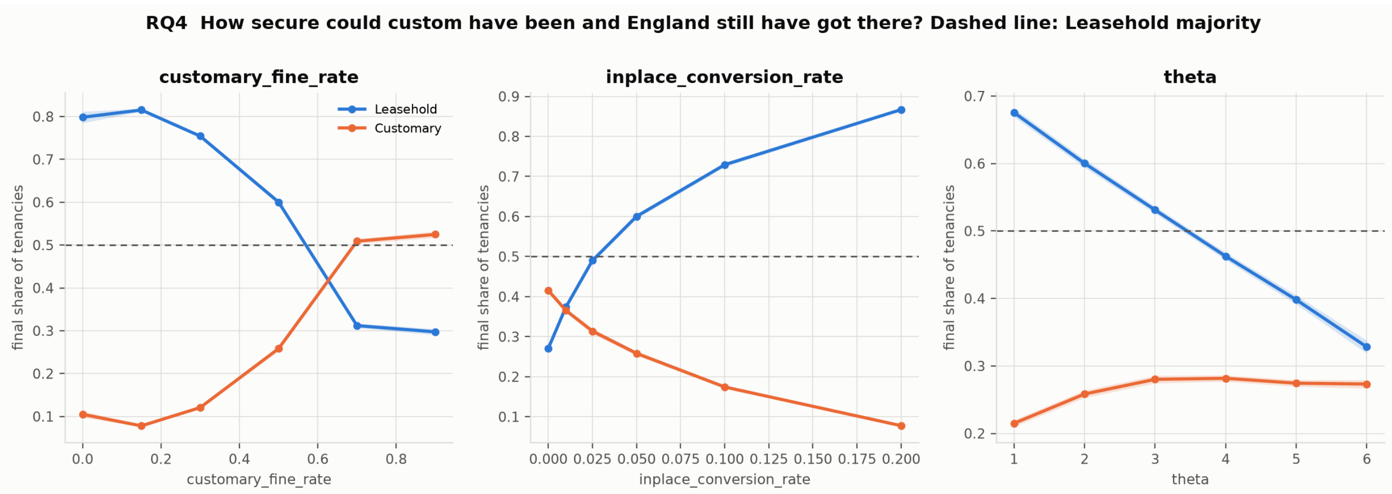

Figure 11: Final Leasehold and Customary shares over the sweep of the arbitrary-fine rate φ, the terms-reopening hazard ϖ and institutional resistance θ, each varied alone. The dashed line is the completion criterion, a Leasehold majority.

Two of the three margins behave as the question anticipates, and one does not. The terms-reopening hazard has a sharp threshold: at $\varpi = 0.025$ the transition completes in 28 of 64 seeds and at $\varpi = 0.05$ in all 64, while at $\varpi \leq 0.01$ it completes in none. Institutional resistance has a threshold in the same sense, completing in 60 of 64 seeds at $\theta = 3$ and in 3 of 64 at $\theta = 4$. On these two margins the bound is straightforward to state: custom could have been about half again as secure as the baseline against the reopening of terms, or about one and a half times as institutionally protected, and England would still have arrived.

The arbitrary-fine rate does not behave monotonically, and the non-monotonicity is a result rather than a nuisance. The final Leasehold share is 0.798 with fines abolished entirely, peaks at 0.815 at $\varphi = 0.15$, and falls thereafter to 0.312 at $\varphi = 0.7$ and 0.297 at $\varphi = 0.9$; the transition completes in all 64 seeds up to $\varphi = 0.5$ and in none from $\varphi = 0.7$. Heavy arbitrary fines *protect* customary tenure in this model. The reason is that the fine is an extractive instrument that substitutes for conversion: a lord able to raise revenue by squeezing a sitting customary tenant has correspondingly less fiscal reason to reopen the tenancy on competitive terms. Custom survives not because it is secure but because it has become lucrative to leave in place. If that is right, then the historiographical inference from evidence of heavy entry fines to evidence of pressure on custom does not go through: within this model the same evidence is equally consistent with custom being farmed rather than dismantled.

Two limits on the frontier should be stated. First, the three margins are swept one at a time, and the two-dimensional group that would have shown their interaction did not complete, so the claim that custom might survive a high reopening hazard *provided* fines stayed low – a claim about the surface, invisible in either margin – remains untested. Given that the fine margin turns out to be non-monotonic, that interaction is more likely to matter than the original cost-based decision to skip it assumed. Second, the frontier is a contour at which the transition stops completing *within the run*, and the horizon group distinguishes that only partially from the run ending too early. Doubling the horizon to 400 periods raises the baseline final share from 0.600 to 0.685 ($+0.085 \pm 0.029$), so the transition is still advancing when the standard run stops. But the arm chosen to represent secure custom in that group ($\varpi = 0.0125$, $\varphi = 0.15$) already completes in all 64 seeds at 200 periods, so it tests the horizon at a point where nothing is censored. The horizon check as run does not establish that the RQ4 frontier is a statement about custom rather than about the length of the run, and re-running it at a configuration that fails at 200 periods – $\varpi = 0$, or $\theta = 5$ – is the outstanding test.

5.6 RQ5: necessity of dispossession

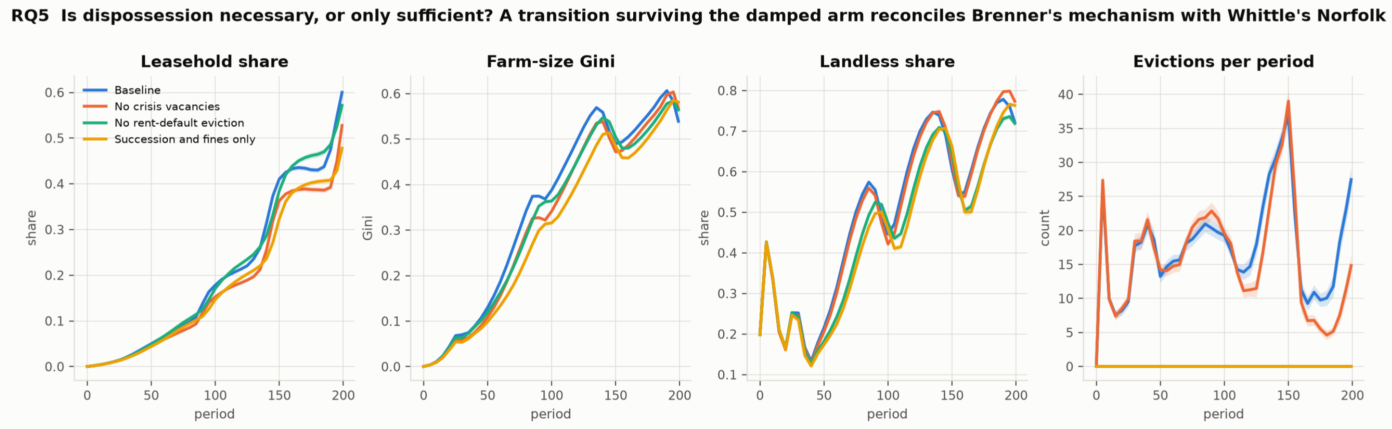

Figure 12: Conversion routed through succession vacancies and fine escalation alone, with the crisis channel damped ($\eta_1 = \eta_2 = 0$, eviction suspended), against the full baseline and against each channel removed singly.

Dispossession is sufficient but not necessary. Damping it to succession and fines alone – no crisis vacancies, eviction suspended for any run of realistic length – costs 0.122 ± 0.011 of final Leasehold share and still transitions 20 of 64 seeds at a mean of 0.478. The individual channels rank as the mechanism predicts but with smaller magnitudes than the account implies: removing crisis vacancies costs 0.073 ± 0.011 and removing rent-default eviction 0.029 ± 0.010, and the fourth panel of Figure 12 shows evictions falling to exactly zero in the damped arms without the transition following them down.

The distributive outcomes are the more striking half of the result, because they do not weaken at all. The damped arm ends with a farm-size Gini of 0.581 against the baseline's 0.539 – higher, by $+0.042 \pm 0.007$ – and a landless share of 0.619 against 0.613, a difference the sign test cannot distinguish from zero ($p = 0.17$). Concentration and a wage-labour pool of the same size are reached without the dispossessive channel that is supposed to produce them, by succession and fine escalation working more slowly on the same land. This is the reconciliation the question was posed to test: Brenner's mechanism can be the route by which England in fact arrived without being the only route the class structure admits, and Whittle's Norfolk evidence of transfer by inheritance and sale rather than by eviction is not a counter-example to the model's account of how customary tenure is extinguished.

5.7 RQ6: ecology or class structure

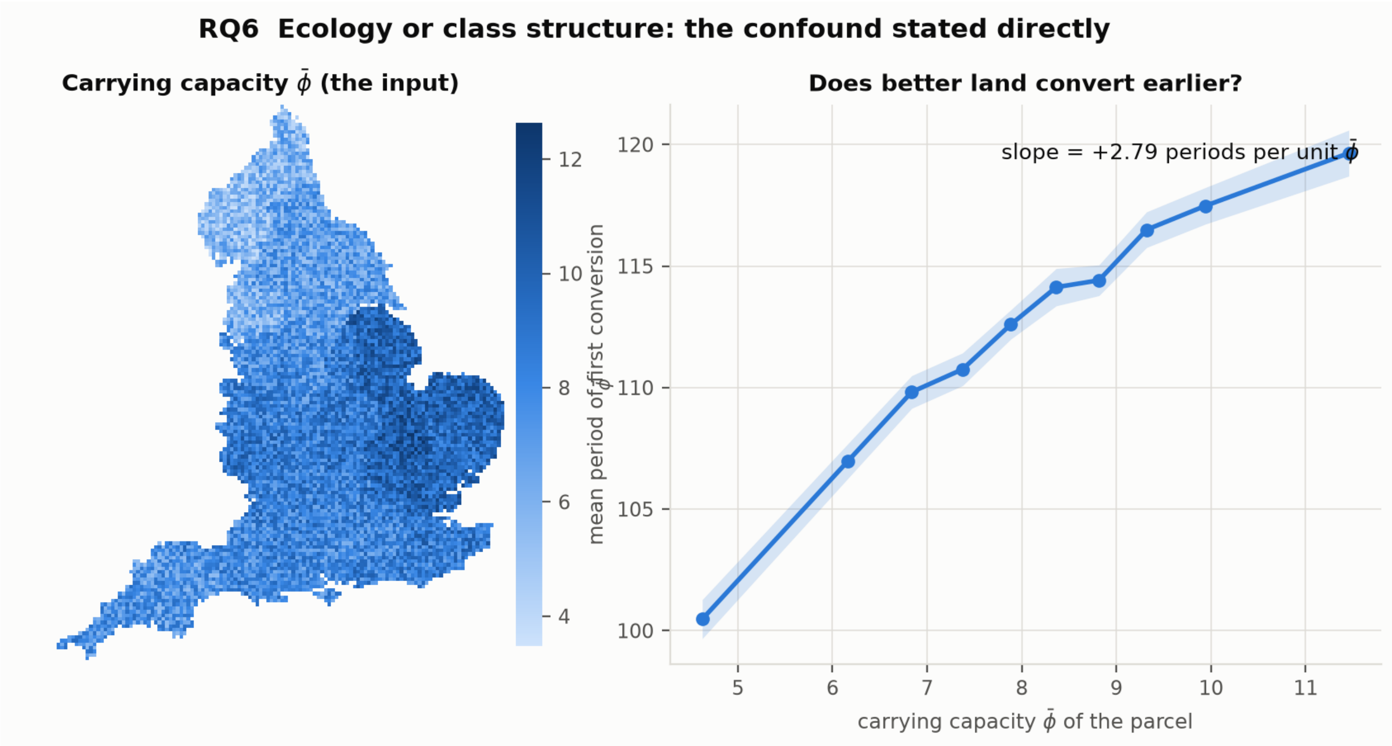

Figure 13: The confound stated directly. Left: the ALC-derived carrying capacity $\bar{\phi}$ that the lattice is built from. Right: mean period of first conversion against that carrying capacity, binned by decile of $\bar{\phi}$.

Ecology sets the order of conversion but not its outcome, and the two halves of that claim need separating. The order is unambiguous and runs opposite to the intuitive direction: conversion arrives *earlier* on poorer land, at a mean first-conversion period of 100.5 in the lowest decile of carrying capacity against 119.6 in the highest, a slope of $+2.79$ periods per unit of $\bar{\phi}$ with intervals nowhere near overlapping. Poor land is where customary rents are eroded to unviability soonest, so it is where the fiscal case for reopening terms arrives soonest; good land sustains the customary arrangement longer precisely because it sustains the customary rent longer. Any account that reads early enclosure as a signature of commercially attractive land has the sign the wrong way round in this model.

The outcome, in contrast, is nearly indifferent to ecology. Flattening the regional fertility structure to its national mean moves the final Leasehold share by $+0.016 \pm 0.008$ – about 2.6% of the baseline level – and additionally suppressing the ecological shock sequence moves it by -0.010 ± 0.008, which the sign test cannot distinguish from zero ($p = 0.10$). Figure 14 shows the ablation parcel by parcel on matching seeds: the difference is a fine-grained speckle of both signs about a national mean of $+0.010$, with no region gaining or losing coherently. What regional fertility does contribute is the small amount of spatial structure the process has at all – flattening it raises the nugget share from 0.889 to 0.908 ($+0.019 \pm 0.005$), which is to say that the little local coherence in conversion timing comes from the land rather than from transmission between lords. That is a point against RQ2's mechanisms rather than for ecology.

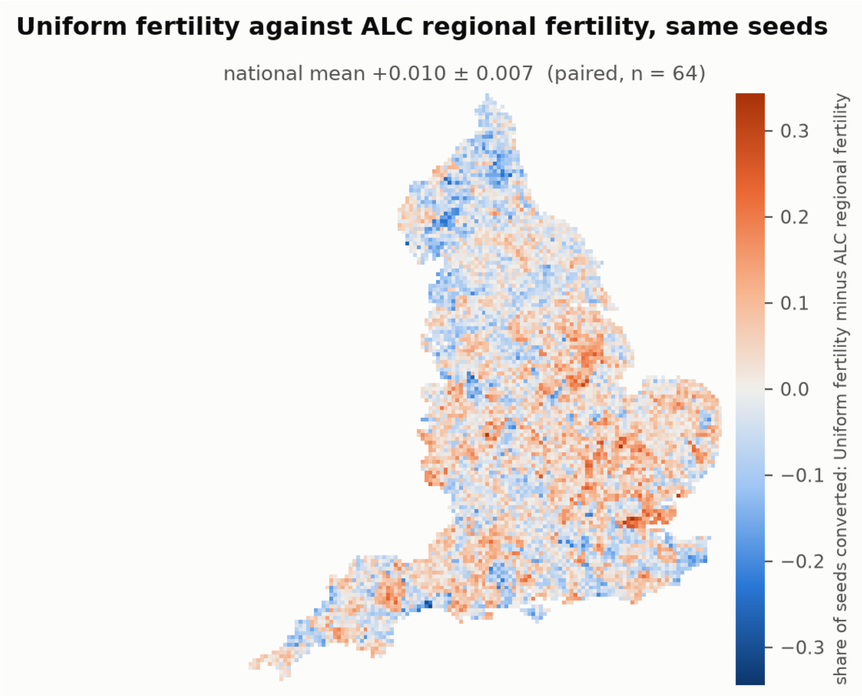

Figure 14: Uniform fertility differenced against ALC regional fertility, parcel by parcel on matching seeds. An ecological driver of the transition would leave a regionally organised difference; this one leaves speckle.

The answer to RQ6 is therefore that ecology is not doing explanatory work the theory attributes to class structure. It schedules the transition – which parcels go first, and about twenty periods of spread between the best and worst land – while the class-structural mechanisms of tiers 3 and 4 determine whether the transition happens at all. Moore's point survives as a claim about sequence and fails as a claim about cause, established here using Brenner's own mechanisms.

5.8 Claims about the model

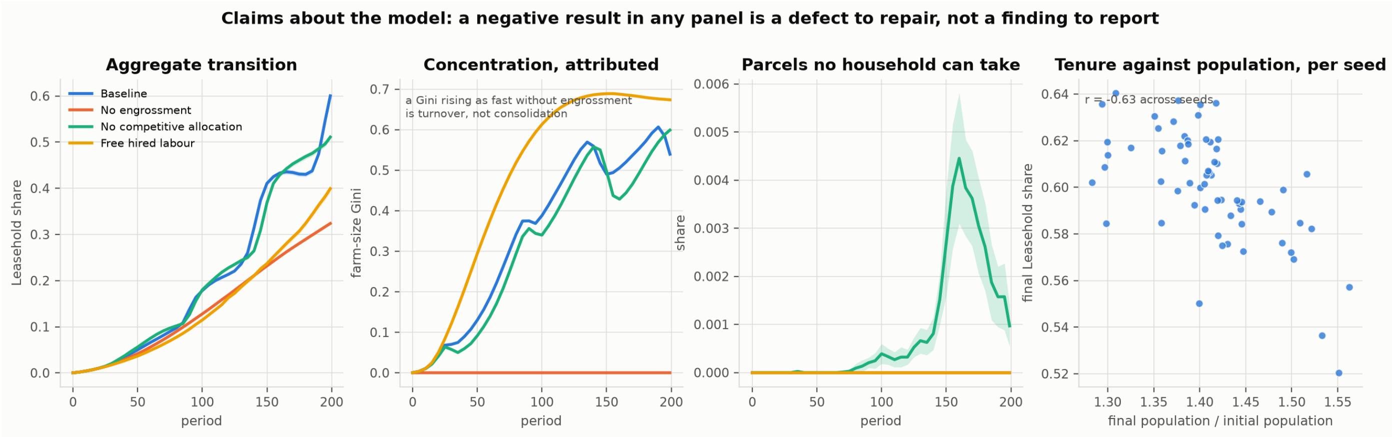

Figure 15: The diagnostics of §??: aggregate tenure composition, the attribution of the farm-size Gini to engrossment rather than turnover, the share of parcels no household can take up, and the cross-seed relation between tenure outcomes and population. A failure in any panel is a defect to repair before the results above can be read.

Three of the five diagnostics pass and two do not, and the two failures constrain how the results above should be read.

Sufficiency of the aggregate transition passes: the baseline reaches a final Leasehold share of 0.600 and transitions in all 64 seeds. *Attribution of concentration* passes decisively. With engrossment disabled the farm-size Gini is exactly 0.000 for the whole run against the baseline's 0.539, so none of the measured concentration is ordinary population turnover; the complementary arm is equally clear, since removing competitive allocation alone *raises* the Gini to 0.599, the entry fine then rationing vacancies by wealth where the auction had rationed them by productivity. *Macro plausibility* passes in the qualitative sense §4.16 claims: hiring continues alongside a growing cumulative exit to industry, and wages and rents diverge in the direction the record suggests, with levels disclaimed throughout.

Population accounting passes on one of its two diagnostics and fails on the other. The share of parcels no household can take up stays at zero in every arm but one, peaking at 0.0045 under the no-competitive-allocation arm, so there is no unmodelled reservoir of prospective tenants. But cross-seed variation in tenure outcomes is *not* independent of cross-seed variation in population: under the baseline household-demography rule the correlation between the final-to-initial population ratio and the final Leasehold share is $r = -0.63$ across the 64 seeds, where the criterion asks for a value near zero. Seeds that grow faster convert less, which is the mechanism identified at tier 6 of the ladder – population growth supplies takers for vacancies that would otherwise be engrossed – operating as a confound rather than as a result. Every share reported above is a share of a determinate total, so the accounting claim holds in its narrow form; but the model's demography and its tenure outcome are coupled, and a reader should not treat a tenure difference between arms as independent of any demographic difference between them.

Independence from the demographic regime fails outright, and it is the most consequential of the two. The final Leasehold share is 0.679 under a closed population ($61/64$ transitioned), 0.600 under household demography ($64/64$), and 0.419 under vital rates applied to single bodies ($6/64$) – a spread of 0.26 between rules, larger than the effect of every mechanism tested above except engrossment itself. Against the closed population the two endogenous rules differ by -0.260 ± 0.028 and -0.079 ± 0.025 respectively. The criterion of §?? is that a result holding under only one such rule is not a result about property relations, and by that criterion the *level* of the transition reported throughout is not portable across demographic regimes. What is portable is harder to state from this group alone, because the accounting arms vary the population rule only at the baseline configuration and do not cross it with the ablations; the ranking of mechanisms, which is what most of the conclusions above rest on, is not directly tested for stability across rules. Establishing that ranking is invariant

– by re-running at least the ladder and RQ2 under the vital-rates rule – is the first thing this model needs before the levels reported here can be defended.

Taken together with the two groups that did not complete, the outstanding work is specific: the demographic-regime cross, the two-dimensional RQ4 frontier, the structural-robustness group underwriting RQ2's spatial scale, and a horizon check placed at a configuration that actually fails within 200 periods. None of the four bears on the qualitative findings – the primacy of engrossment, the simultaneity of the transition, the sufficiency-without-necessity of dispossession, the scheduling role of ecology – and all four bear on the numbers attached to them.

6 Discussion

7 Conclusions

8 Acknowledgements

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